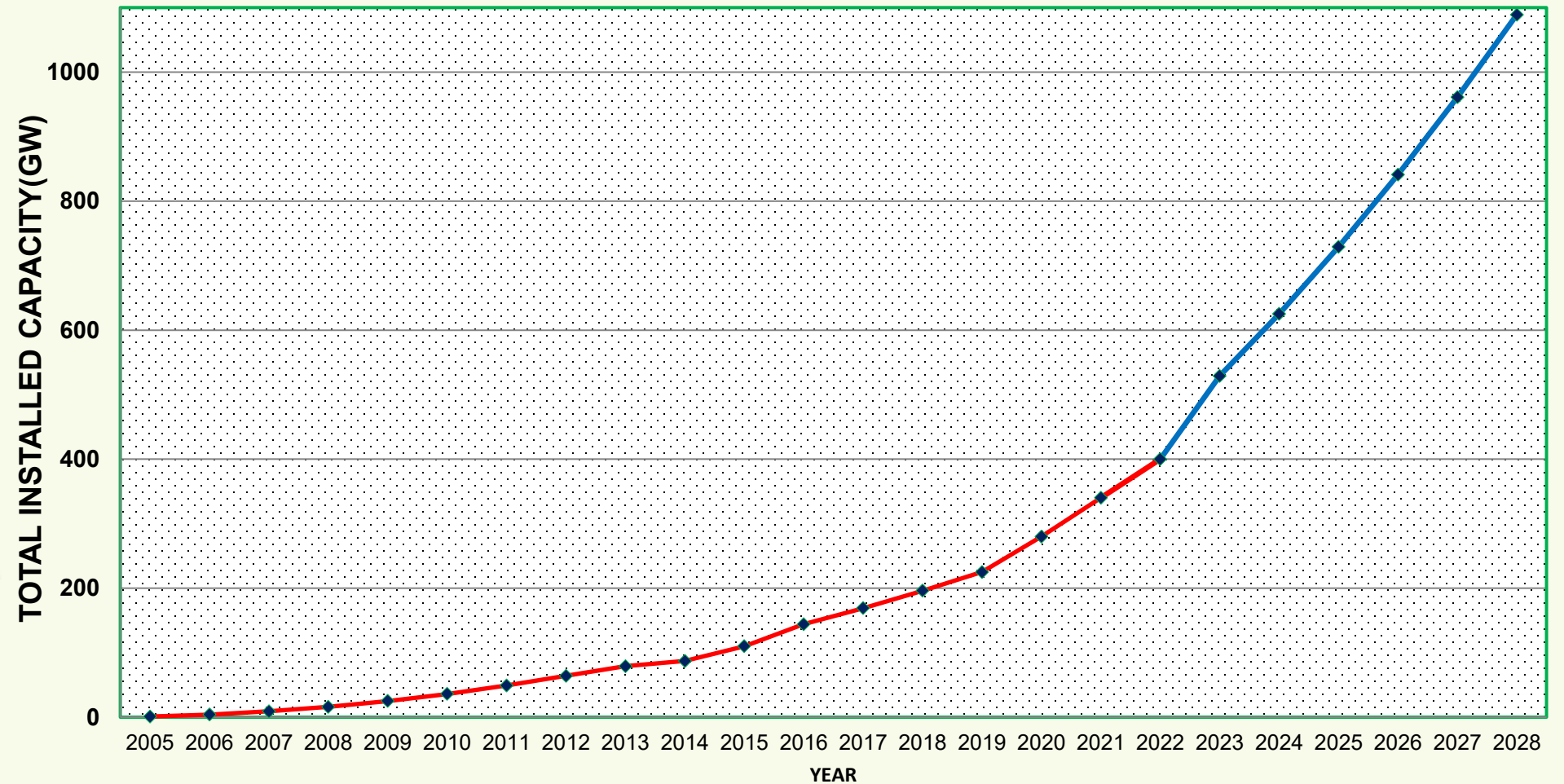


## FORECAST



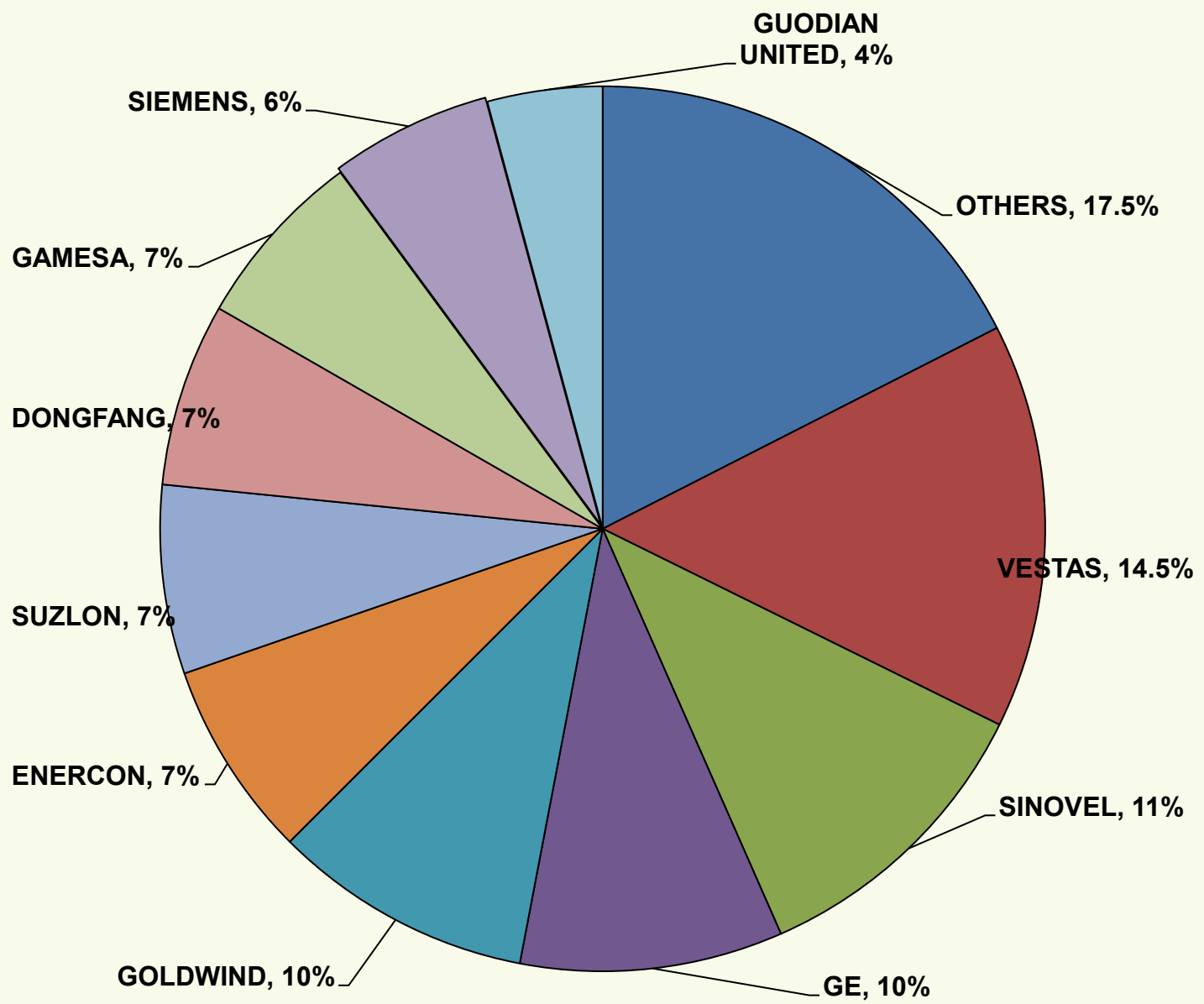
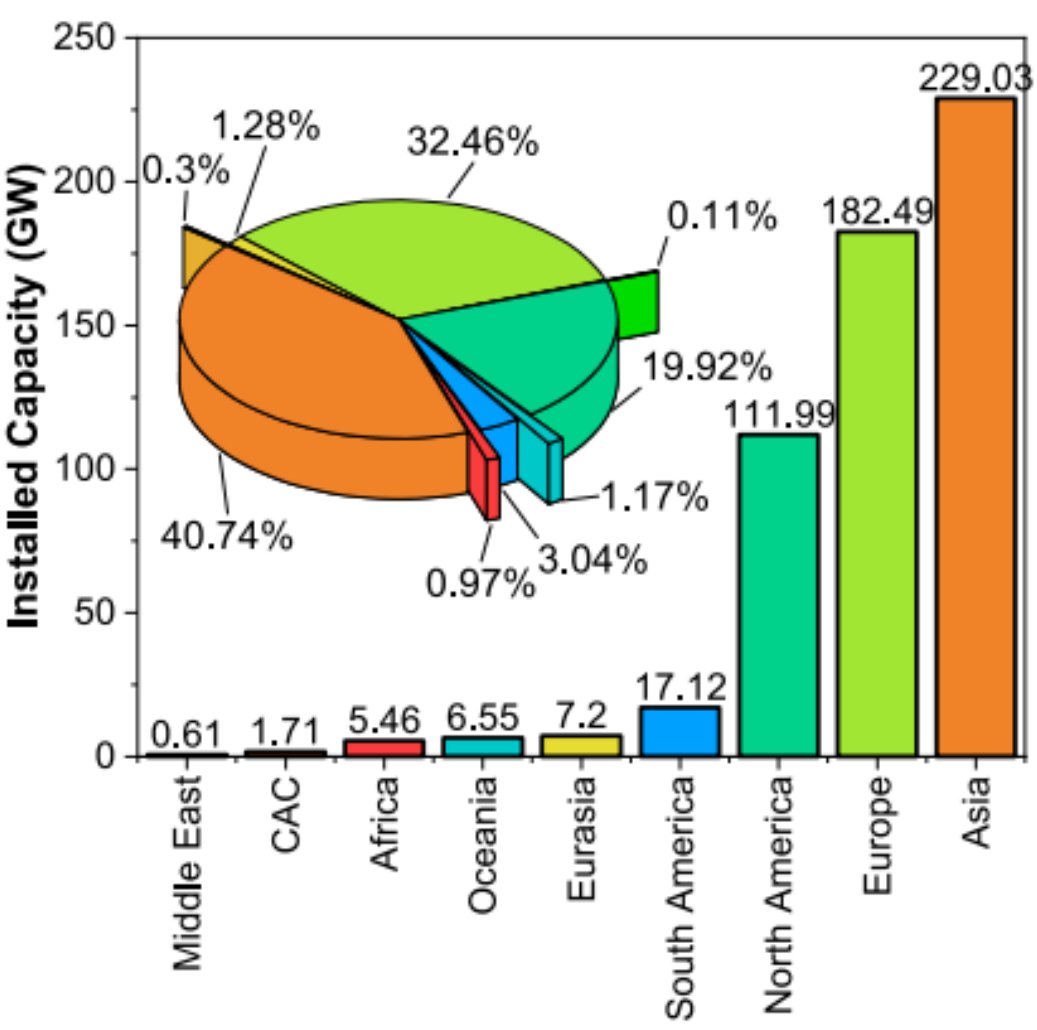
Wind energy global capacity (**red**) and forecast (**blue**) [1]

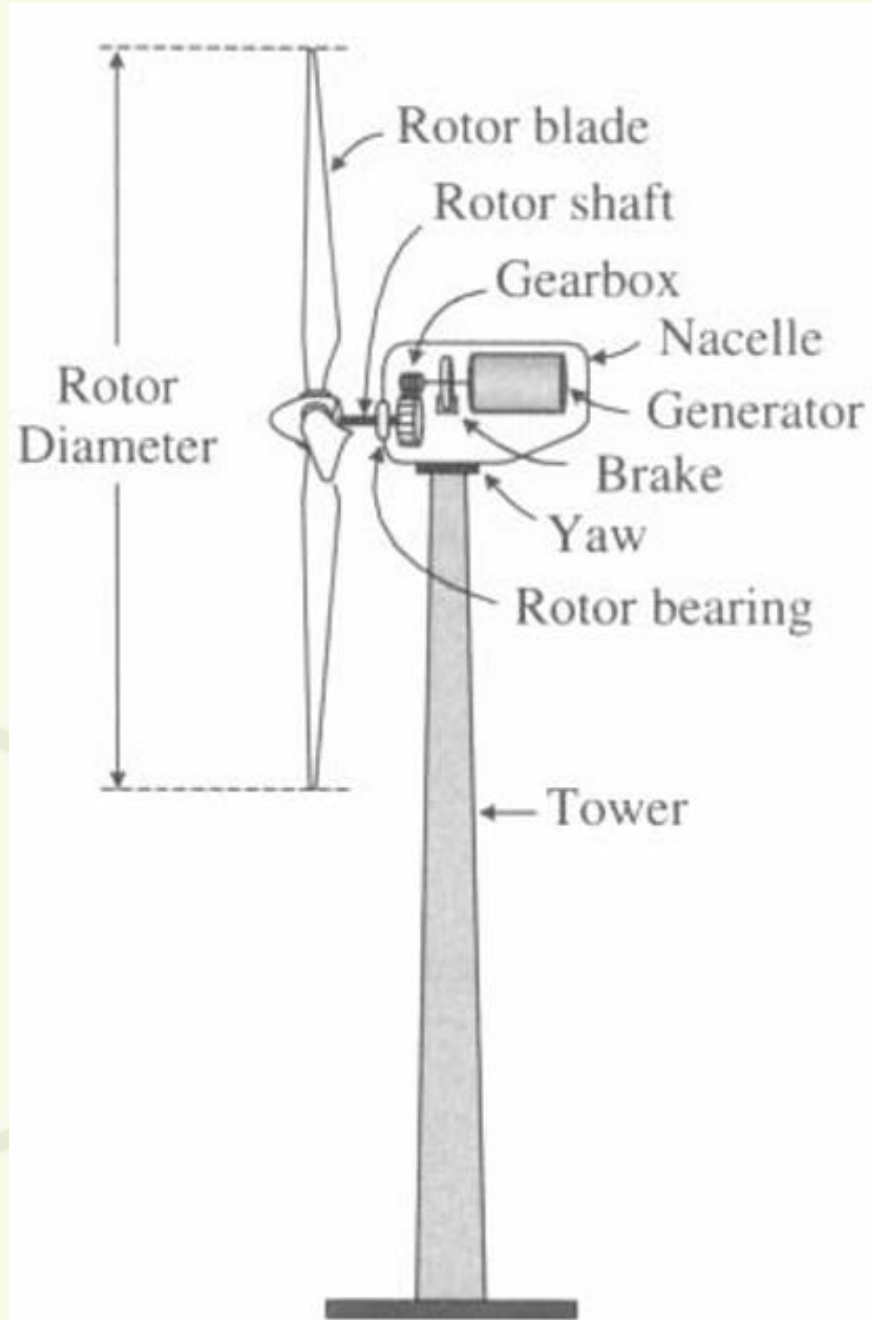
In 2005s (500 kW) to 2023 (567 GW)

Tower heights

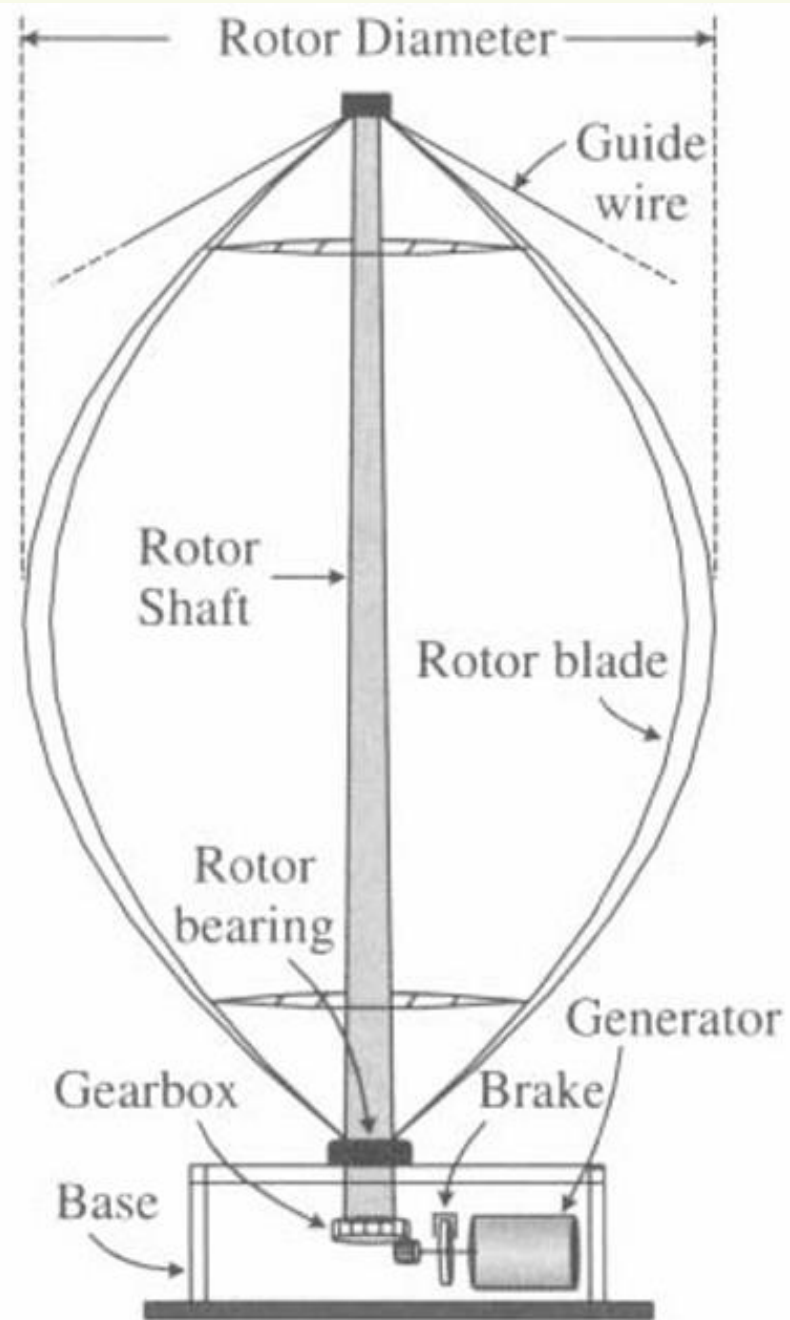
Rotor diameters increased

Overall weight, size and capacity increased





(a) Horizontal-axis turbine



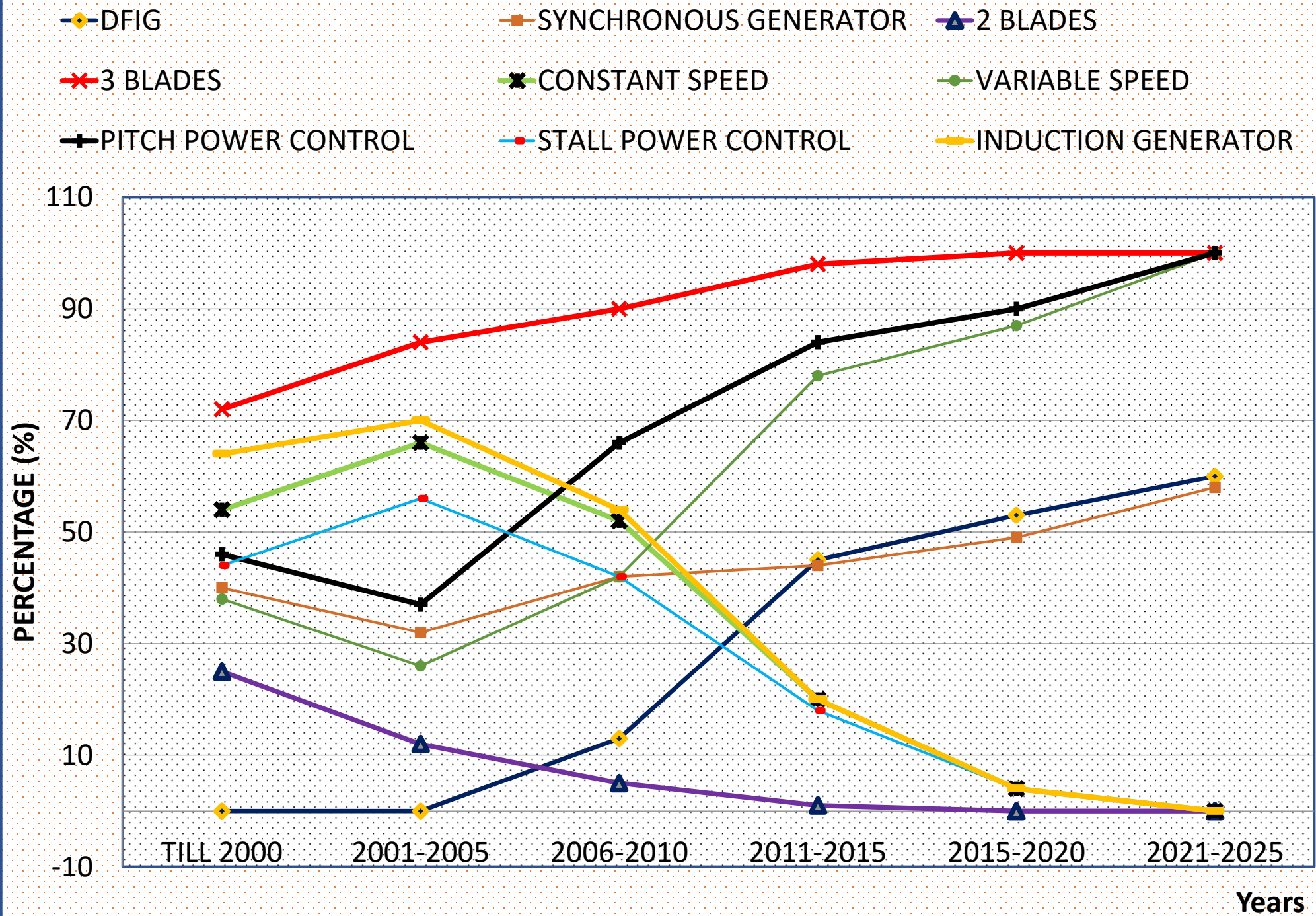
(b) Vertical-axis turbine

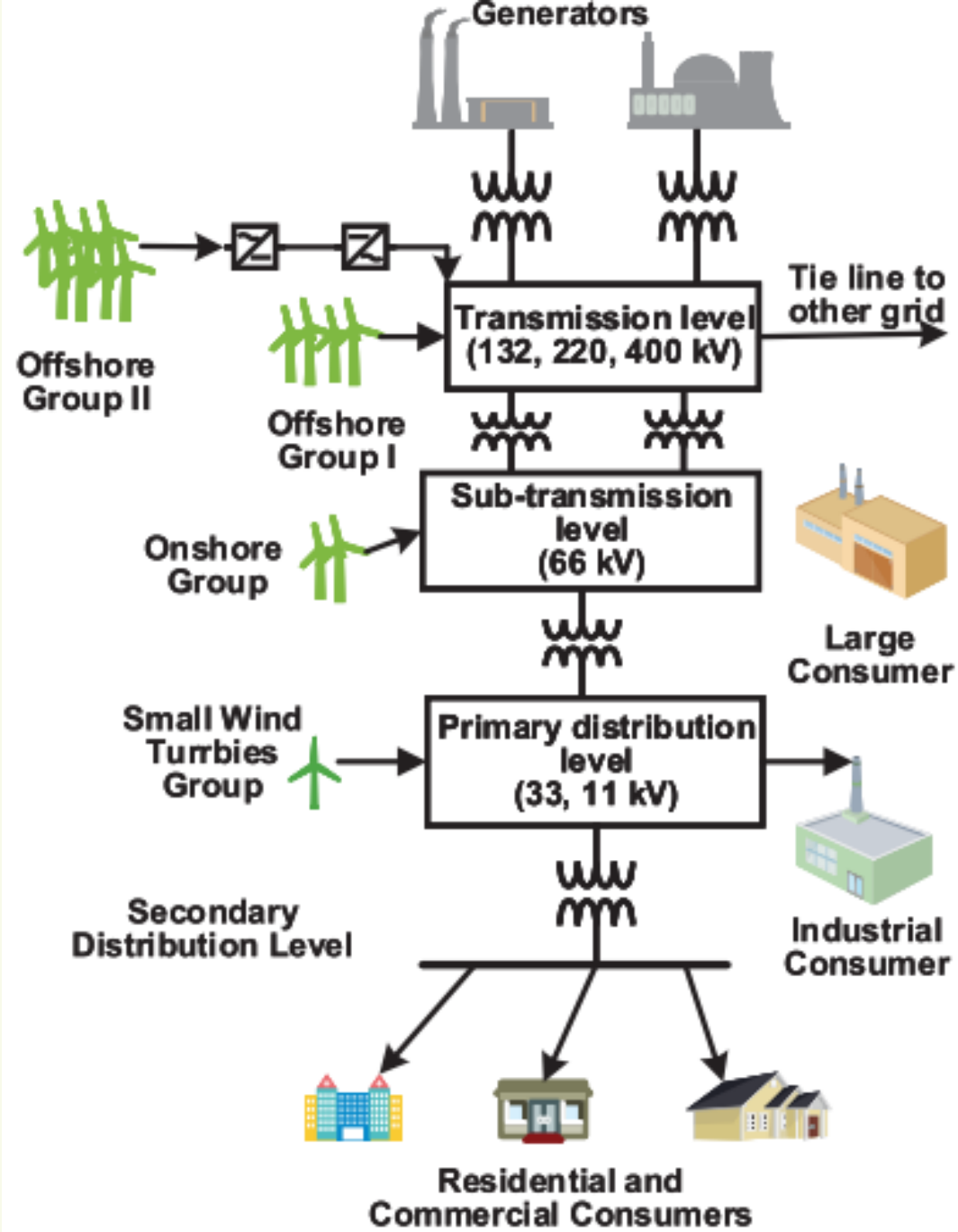
Table 1 Comparison between horizontal- and vertical-axis wind turbines

| Turbine type | Advantages                                                                                                                                                                                                                                                                 | Disadvantages                                                                                                                                                                                                                            |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HAWT         | <ul style="list-style-type: none"> <li>• Higher wind energy conversion efficiency</li> <li>• Access to stronger wind due to high tower</li> <li>• Power regulation by stall and pitch angle control at high wind speeds</li> </ul>                                         | <ul style="list-style-type: none"> <li>• Higher installation cost, stronger tower to support heavy weight of nacelle</li> <li>• Longer cable from the top of tower to ground</li> <li>• Orientation required (yaw control)</li> </ul>    |
| VAWT         | <ul style="list-style-type: none"> <li>• Lower installation cost and easier maintenance due to the ground-level gearbox and generator</li> <li>• Operation independent of wind direction</li> <li>• Suitable for rooftops (stronger wind without need of tower)</li> </ul> | <ul style="list-style-type: none"> <li>• Lower wind energy conversion efficiency</li> <li>• Higher torque fluctuations and prone to mechanical vibrations</li> <li>• Limited options for power regulation at high wind speeds</li> </ul> |

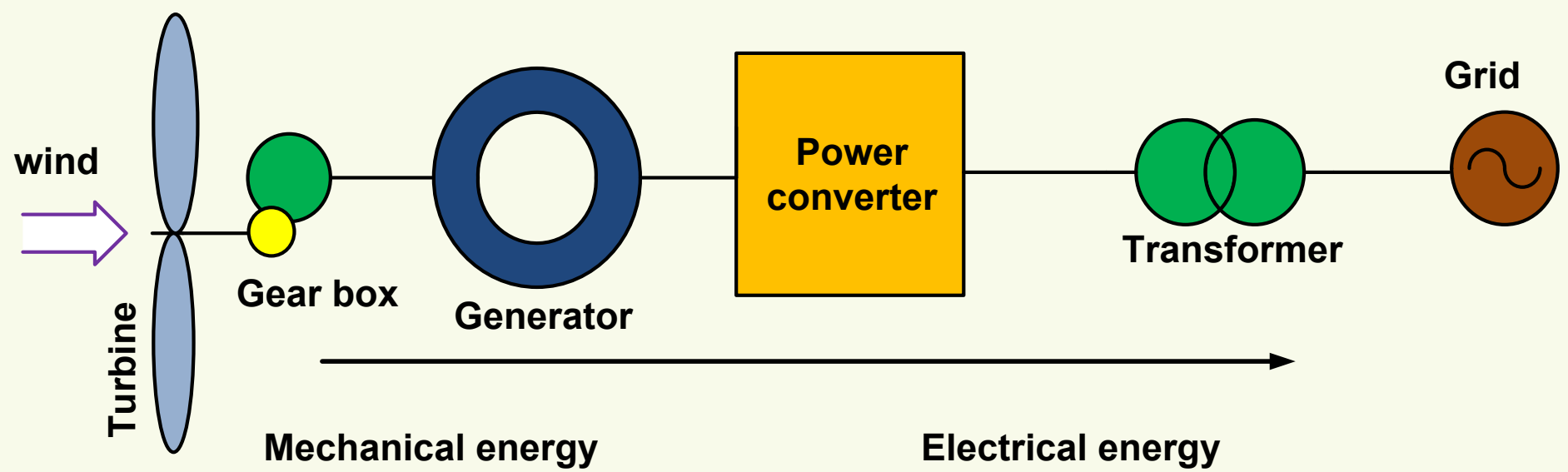
Table Advantages and drawbacks of fixed- and variable-speed wind turbines

| Speed mode     | Advantages                                                                                                                                               | Disadvantages                                                                                                                                                                |
|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fixed speed    | <ul style="list-style-type: none"><li>• Simple, robust, reliable</li><li>• Low cost and maintenance</li></ul>                                            | <ul style="list-style-type: none"><li>• Relatively low energy-conversion efficiency</li><li>• High mechanical stress</li><li>• High power fluctuations to the grid</li></ul> |
| Variable speed | <ul style="list-style-type: none"><li>• High-energy conversion efficiency</li><li>• Improved power quality</li><li>• Reduced mechanical stress</li></ul> | <ul style="list-style-type: none"><li>• Additional cost and losses due to use of converters</li><li>• More complex control system</li></ul>                                  |

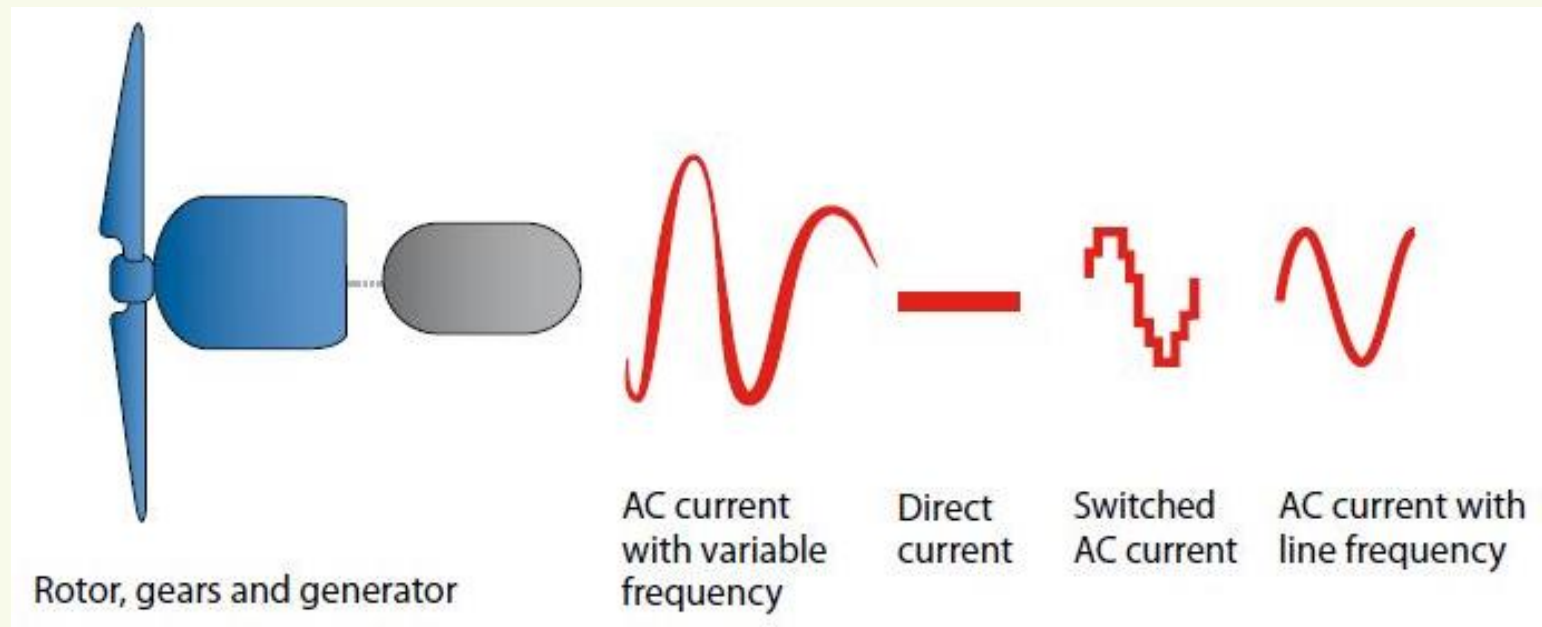




Power system grid with wind integration



Wind energy conversion form mechanical to electrical



Power Electronic Converter (PEC)

# Types of Wind Turbine Generators (WTGs)

The main generators configurations used in WTs are

(A) **Squirrel Cage Induction Generator (SCIG)**

(B) **Wound Rotor Induction Generator (WRIG)**

(C) **Permanent Magnet Synchronous Generator (PMSG)**

or

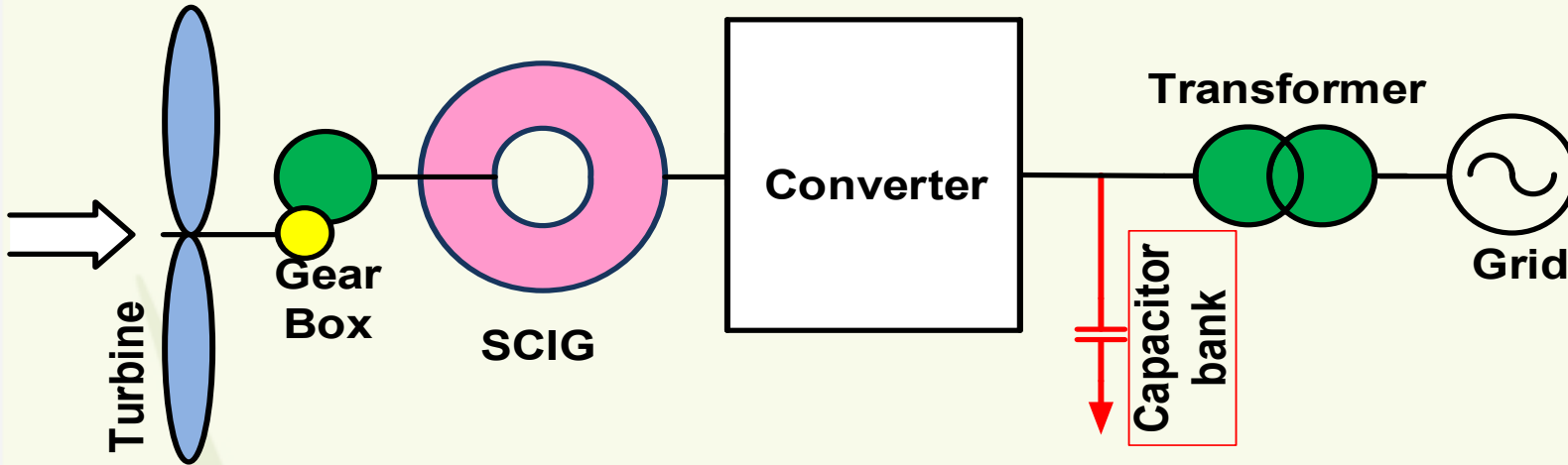
**Electrically Excited Synchronous Generator (EESG)**

or

**Direct-driven Permanent Magnet Wind Power Generators (DDPMG)**

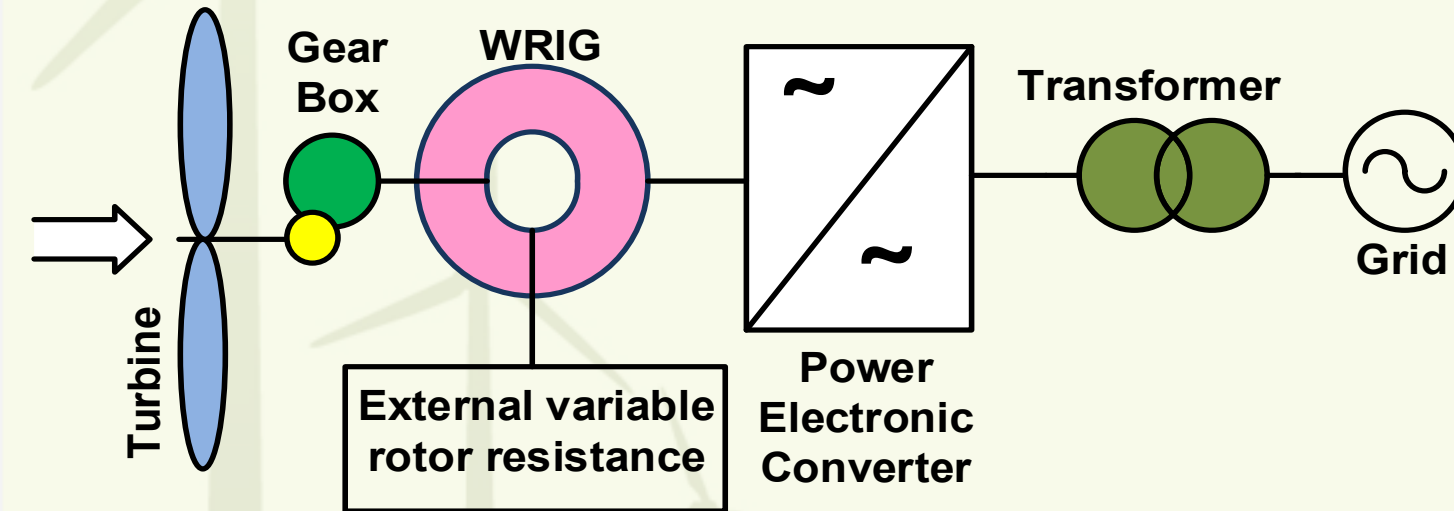
(D) **Doubly Fed Induction Generator (DFIG)**

### (A). Squirrel Cage Induction Generator (SCIG)



Constant speed wind power generation system with SCIG

### (B). Wound Rotor Induction Generator (WRIG)



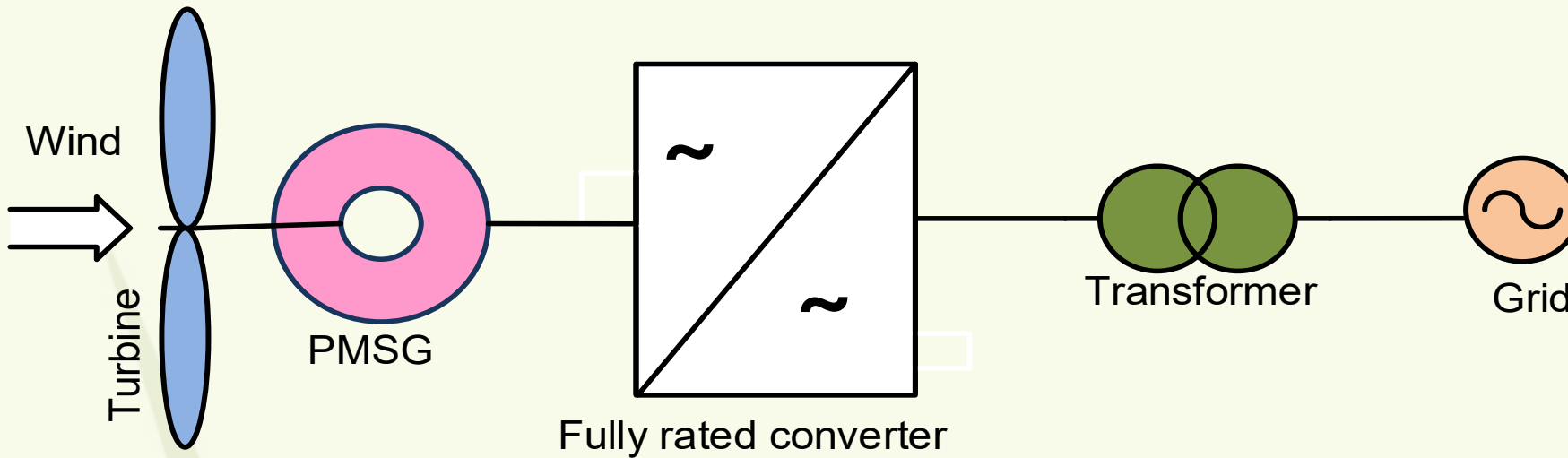
Variable speed wind power generation system with WRIG

### Type A and B System:

- Induction generator (squirrel cage (type A) and wound rotor (type B))
- Capacitors (Provide Reactive power to induction generator and to the grid (very limited))
- Gearbox to boost shaft speed (nearly 100 times)
- Slip rings for type B.
- Control resistor for type B.

Introduces additional **heat dissipation**

### (C). Permanent Magnet Synchronous Generator (PMSG)

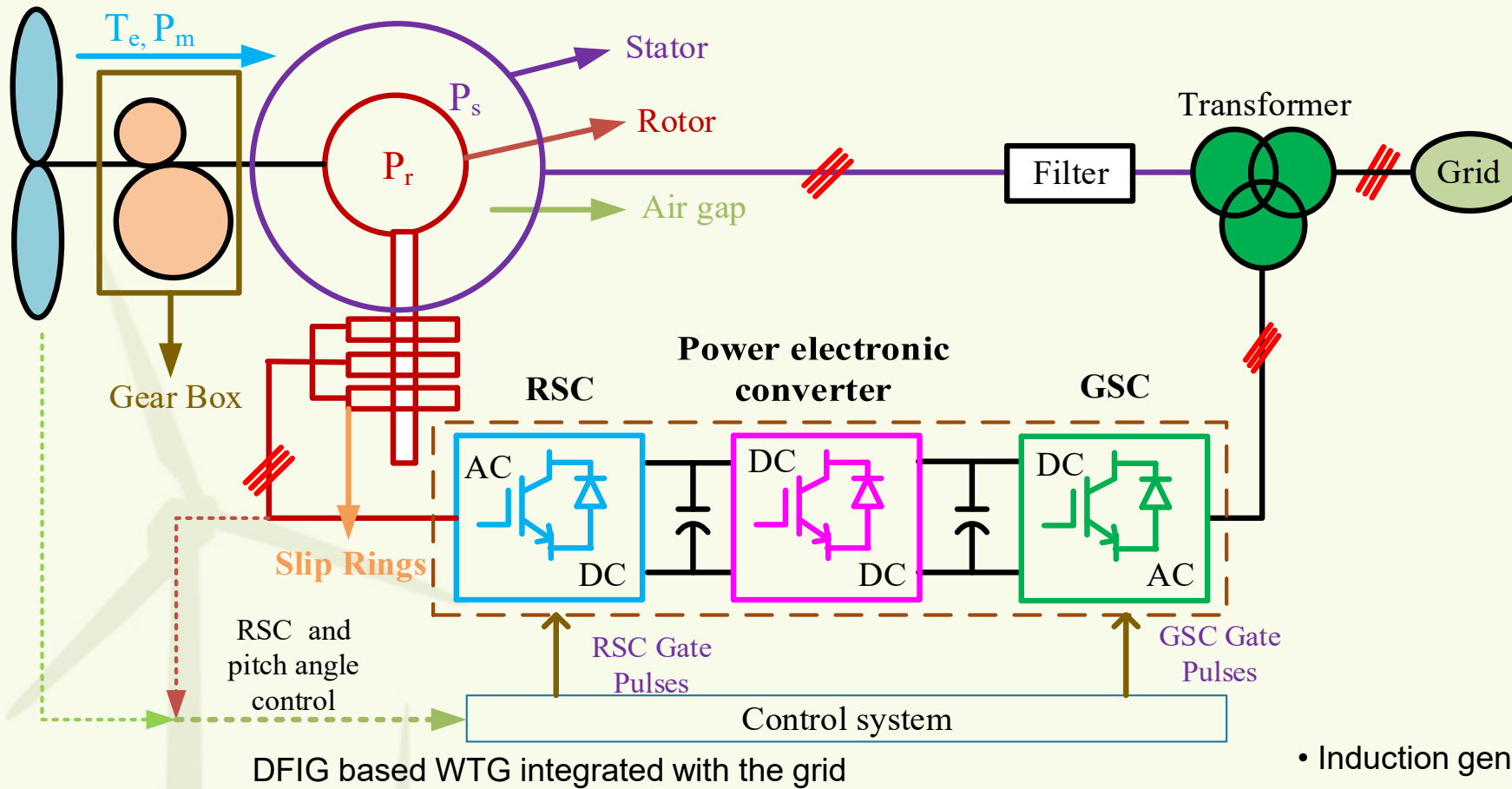


Variable speed wind power generation system with PMSG (fully-rated PEC)

#### Type C System:

- Generator (asynchronous, permanent magnet) or synchronous.
- AC/DC generator side converter (convert AC (different frequency) to DC)
- DC/AC grid side converter convert DC to AC (50 Hz / 60 Hz as needed).
- Gearbox (eliminated in direct drive).

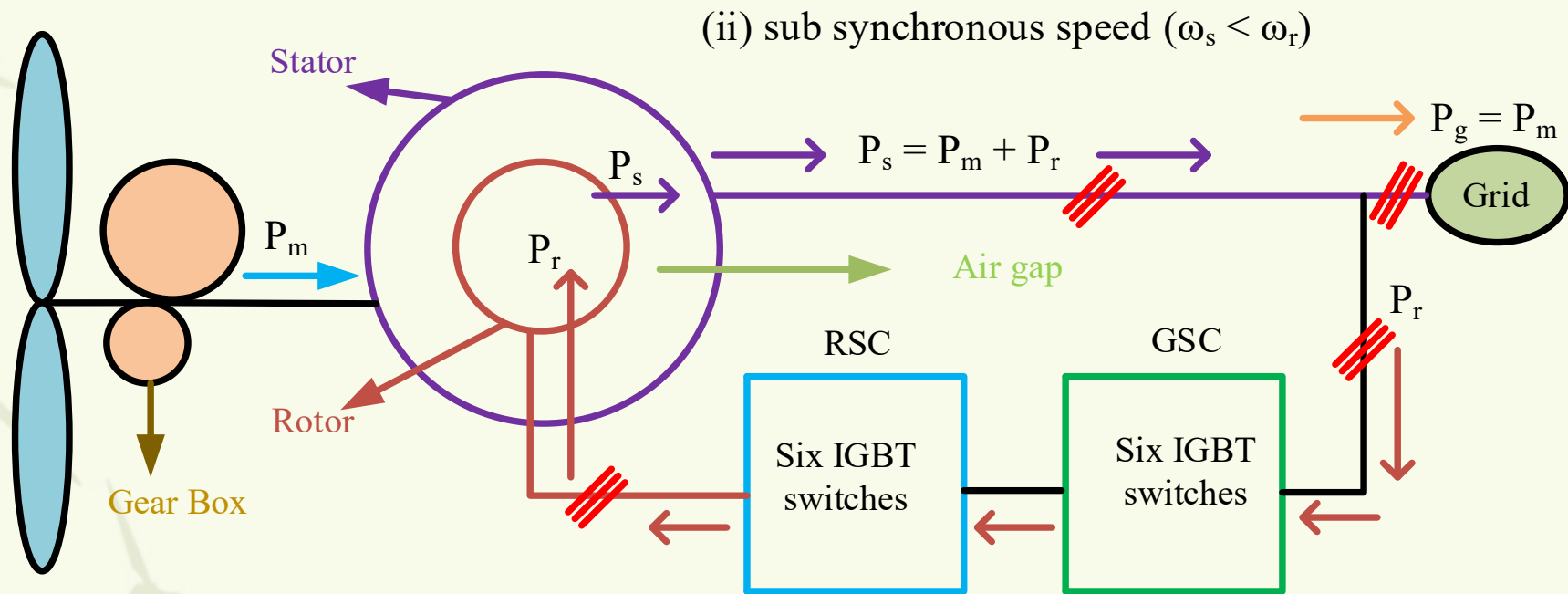
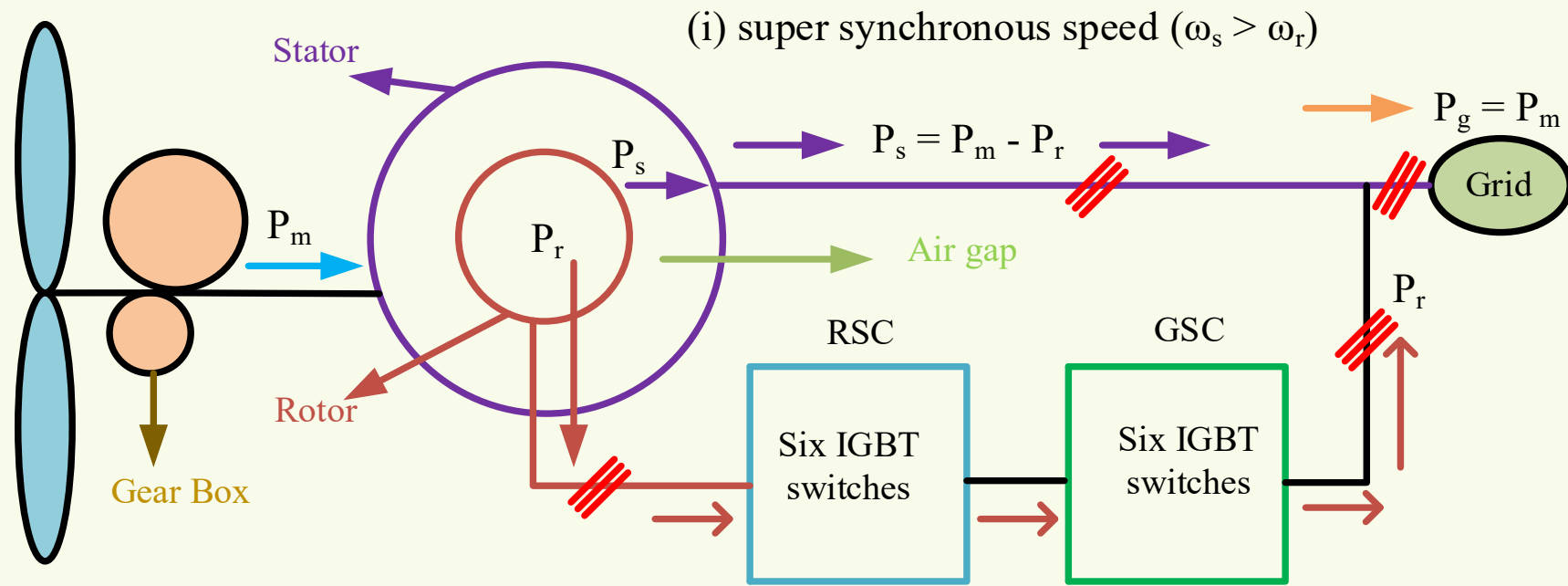
## (D). Doubly Fed Induction Generator (DFIG)



### Type D System:

- Induction generator (wound rotor), double-fed (DFIG).
- Gearbox to boost shaft speed (nearly 100 times).
- AC/DC-DC/AC converter (provide second supply to the wound rotor)
- Crowbar to protect rotor winding throughout the faults.

- The DFIG is essentially a wound rotor induction generator in which the rotor circuit can be controlled by external devices to achieve variable speed operation.
- Leading and lagging reactive power



Power flow directions in DFIG of WTGs.

## Wind turbine evolution and the main trend of power electronic conversion in the last 35 years

| YEAR        | DIAMETER<br>(METER) | POWER<br>GENERATION | POWER ELECTRONIC CONVERSION                                                        |
|-------------|---------------------|---------------------|------------------------------------------------------------------------------------|
| 1990        | 15 m                | 50 KW               | Rating is 0%, and Operational role is <b>Soft starter<br/>(A Type)</b>             |
| 1995        | 20 m                | 100 KW              |                                                                                    |
| 2000        | 40 m                | 500 KW              |                                                                                    |
| 2000        | 50 m                | 600 KW              | <b>Rating is 10%, and Operational role Rotor<br/>resistance control (B Type)</b>   |
| 2005        | 80 m                | 2 MW                |                                                                                    |
| 2010 - 2025 | 124 m               | 5 MW                | Rating is 30%, and Operational role is the <b>Rotor<br/>power control (C Type)</b> |
| 2015 - 2025 | 164 m               | 7-7.5 MW            | Rating is 100% and Operational role is Full<br>generator power control (D Type)    |
|             | 210 m               | 10 MW               |                                                                                    |

# Summary of Wind Energy Conversion Systems

| Generator<br>Concept (Type)               | SCIG (A)                                                                                            | WRIG (B)                                               | PMSG(C)                                                                                                 | DFIG (D)                                                                                                             |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------------------------------------------|---------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Type of speed                             | Fixed/constant speed wind turbine                                                                   | Limited variable speed wind turbine                    | Variable speed wind turbine                                                                             | Variable speed wind turbine                                                                                          |
| Gearbox stages                            | 3                                                                                                   | 3                                                      | None                                                                                                    | 3                                                                                                                    |
| Control                                   | Active stall, Passive stall Pitch                                                                   | Pitch                                                  | Pitch                                                                                                   | Pitch                                                                                                                |
| Individual speed control                  | Yes                                                                                                 | No                                                     | No                                                                                                      | Yes                                                                                                                  |
| Power output of the system control by     | Soft starter                                                                                        | Rotor resistance control                               | Full generator power control                                                                            | Rotor power control                                                                                                  |
| Energy produced                           | Less                                                                                                | Moderate-high                                          | High                                                                                                    | Moderate-high                                                                                                        |
| Standby function                          | Yes                                                                                                 | No                                                     | Yes                                                                                                     | Yes                                                                                                                  |
| Reliability                               | Very high                                                                                           | High                                                   | High                                                                                                    | Moderate                                                                                                             |
| Technical growth                          | High                                                                                                | High                                                   | Medium- high                                                                                            | High                                                                                                                 |
| Electric grid support for stability       | Low                                                                                                 | High                                                   | High                                                                                                    | Medium                                                                                                               |
| Power ratings (kW) And Rotor Diameter (m) | (270, 27)/<br>(300, 28)/<br>(600, 43)/<br>(800, 49)/<br>(1000, 54)/<br>(1320, 61)/<br>(2300, 82.4). | (600, 39)/<br>(660, 47)/<br>(1800, 80)/<br>(2100, 88). | (600, 40)/<br>(1500, 70)/<br>(2500, 90)/<br>(3000, 101)/<br>(3600, 107)/<br>(4100, 113)/<br>(4500, 114) | (1500,70.4/77/82)/<br>(1800, 90)/<br>(2500, 90)/<br>(2250, 88)/<br>(2750, 80)/<br>(2000, 90/92.5)/<br>(3000, 90/91.6 |

| Cost                          | Less                                                                                                                                                                                 | High                                                                                                                                                                                            | Medium-high                                                                                                                                                                                                                                | Medium                                                                                                                                                                                                                                     |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Maintenance                   | Medium                                                                                                                                                                               | High                                                                                                                                                                                            | Medium                                                                                                                                                                                                                                     | Medium                                                                                                                                                                                                                                     |
| Reactive power control        | <ul style="list-style-type: none"><li>• Capacitor banks</li><li>• SVCs or SATCOM</li><li>• No frequency control</li></ul>                                                            | <ul style="list-style-type: none"><li>• Fully scale frequency/Power converter</li></ul>                                                                                                         | <ul style="list-style-type: none"><li>• Fully scale frequency/Power converter</li></ul>                                                                                                                                                    | <ul style="list-style-type: none"><li>• Partially/fully scale frequency/power converter (25-30%)</li><li>• For some special case extra protection required for a crowbar</li></ul>                                                         |
| Advantages                    | <ul style="list-style-type: none"><li>• Easier to design, construct and control</li><li>• Robust operation</li></ul>                                                                 | <ul style="list-style-type: none"><li>• High energy yield</li><li>• Higher active/reactive Power controllability</li><li>• Absence of brush/slip ring</li><li>• Low mechanical stress</li></ul> | <ul style="list-style-type: none"><li>• Highest energy yield</li><li>• Higher active/reactive Power controllability</li><li>• Absence of brush/slip ring</li><li>• Low mechanical stress</li><li>• No copper losses on the rotor</li></ul> | <ul style="list-style-type: none"><li>• High energy yield</li><li>• High active/reactive power Controllability</li><li>• Lower cost on PEC</li><li>• Lower losses by PEC</li><li>• Less mechanical stress</li><li>• Compact size</li></ul> |
| Disadvantage                  | <ul style="list-style-type: none"><li>• Low energy yield</li><li>• No active/reactive power controllability</li><li>• High mechanical stress</li><li>• High losses on gear</li></ul> | <ul style="list-style-type: none"><li>• Higher cost of copper winding</li><li>• Higher cost on PEC</li><li>• Higher losses on PEC</li><li>• Large size</li></ul>                                | <ul style="list-style-type: none"><li>• High cost of PM material</li><li>• Demagnetization of PM</li><li>• Complex construction process</li><li>• Higher cost on PEC</li><li>• Higher losses on PEC</li><li>• Large size</li></ul>         | <ul style="list-style-type: none"><li>• Existence of brush/slip ring</li><li>• High losses on gear</li></ul>                                                                                                                               |
| Major manufacturing companies | Vestas (V27),<br>Siemens (B54, B82),<br>Neg Micon (M530, M1500),<br>Tacke (TW600),<br>Made (AE-61),<br>Nordex (N52),                                                                 | Vestas (39,44,47),<br>Suzlon(S88),<br>Gamesa (G80).                                                                                                                                             | GE (GE 4,1-113),<br>Goldwind (GW70,90),<br>Enercon (E-40,66,112, 126),<br>Dongfang (DF82-1500),<br>Gamesa (G128),<br>Made (D13W),                                                                                                          | Vestas (V90),<br>Sinovel (1500,3000),<br>GE(GE77),<br>Suzlon(S88-MarkII),<br>Gamesa(G90),<br>Guardian (U.P 1500),<br>NagMicon (NM80),                                                                                                      |

Integrated wind farm when compared to conventional synchronous generator under transient behavior



synchronous generators accelerate or decelerate



generator rotor speed and grid frequency changes



Inertia plays a role in suppressing the rapid changes in rotor speed and electrical power output



Wind farm are generally, back-to-back converters provide a fast and accurate variable-frequency excitation ( $\pm 30\%$ ) or fully rated converter

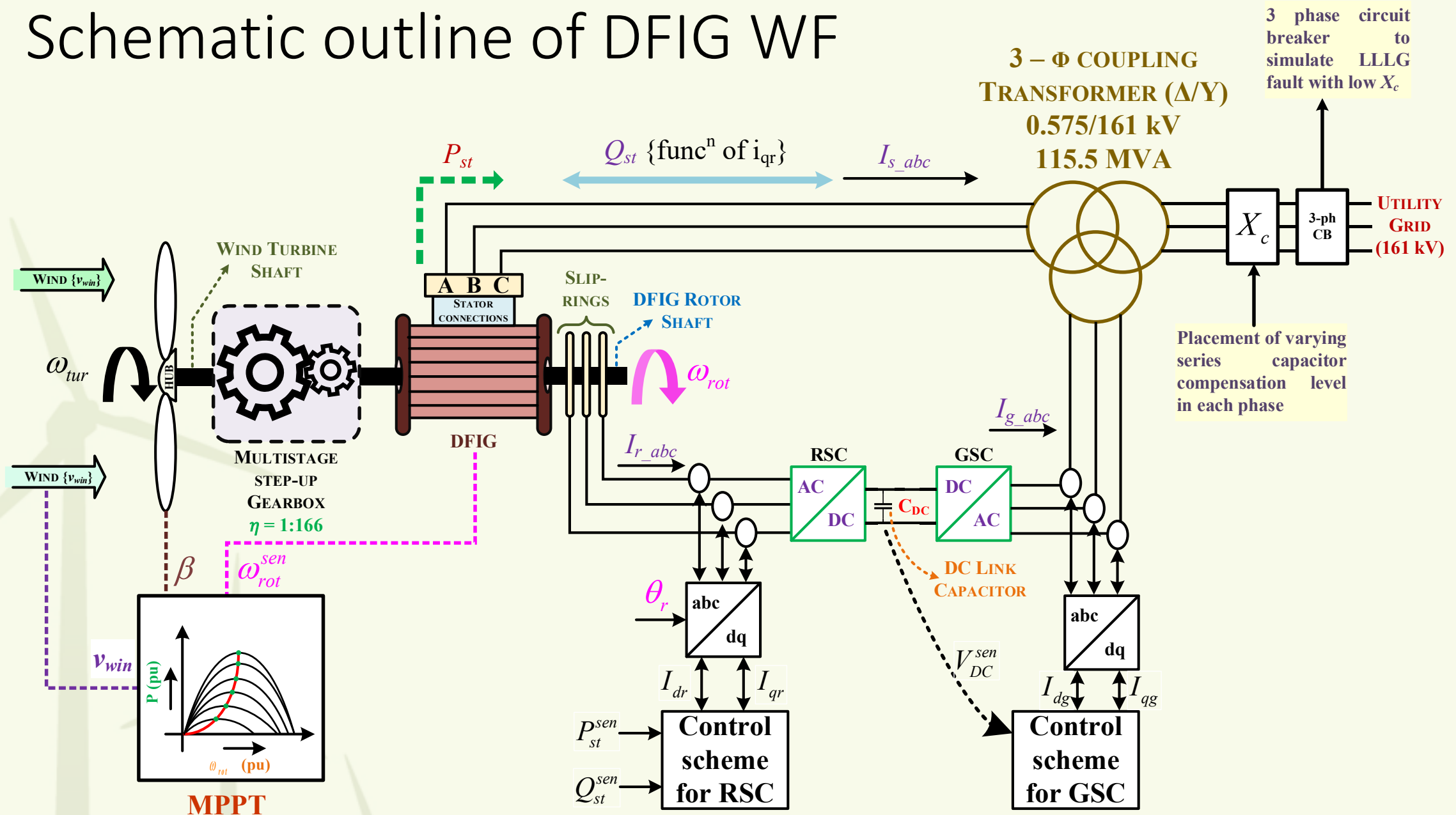


No concepts of “relative swing”, “rotor angle deviation”, “power angle characteristic”, and no inertia concept



It suffers from Low Voltage Ride through (LVRT)

# Schematic outline of DFIG WF



Schematic of DFIG WF

# Design of the wind turbine and gear mechanism

$$P_m = \frac{1}{2} C_p \rho_{air} A_s (v_w)^3 = \frac{1}{2} C_p \rho_{air} \pi r_{wt}^2 (v_w)^3$$

whereas  $P_m \rightarrow$  **mechanical wind power of an air mass (1.5 MW)**

$v_w \rightarrow$  **wind velocity (9 m/s)**

$A_s \rightarrow$  **swept area:** area occupied by rotating wind blades in  $m^2$

$C_p \rightarrow$  **power coefficient:**  $f(\beta, \lambda)$   
(0.3755)

$\rho_{air} \rightarrow$  **specific air density (1.1644 kg/m<sup>3</sup> at 30°C)**

$r_w \rightarrow$  **radius of wind turbine (54.735 m)**

**Pitch angle = 0 (since wind speed is constant)**

*Pitch angle=0*

*Tip speed ratio*

# Design of the wind turbine and gear mechanism

$$C_p = 0.5176 \left( \frac{116}{\lambda_i} - 0.4\beta - 5 \right) e^{-\frac{21}{\lambda_i}} + 0.0068\lambda$$

$$\text{where, } \frac{1}{\lambda_i} = \frac{1}{\lambda + 0.08\beta} - \frac{0.035}{\beta^3 + 1}$$

$$C_p = 0.3755$$

Relationship between linear speed of turbine blade's tip and wind speed

$$\begin{aligned} \lambda &= \frac{\text{tip speed of the blade}}{\text{wind speed}} = \frac{v_{tip}}{v_w} \\ &= \frac{\omega_{wt} \times r}{v_w} = \frac{2\pi N_{wt} \times r}{60v_w} \end{aligned}$$

$$\lambda = 6$$

Gear ratio

$$\eta = \frac{\omega_r}{\omega_{wt}}$$

$$\eta = 1:166$$

# Calculation for stator and rotor powers (formulas)

$$J_{total} \frac{d}{dt} \omega_r = \tau_m - \tau_{em} - F \omega_r$$

$J_{tur} + J_{DFIG}$

Eqn. 1

$$P_m = \tau_m \omega_r$$

Eqn. 2

$$P_s = \tau_{em} \omega_s$$

Eqn. 3

$$P_m = P_t$$

Eqn. 4

$$P_t = P_s + P_r$$

Eqn. 5

$$P_r = P_t - P_s$$

Eqn. 6

$$P_r = \tau_m \omega_r - \tau_{em} \omega_s$$

Eqn. 7

$$P_r = -\tau_{em} \left( \frac{\omega_s - \omega_r}{\omega_s} \right) \omega_s$$

Eqn. 7

$$P_r = -s P_s$$

Eqn. 8

$$P_s = \frac{P_t}{1-s}$$

Eqn. 9

Whereas  $s = \frac{\omega_s - \omega_r}{\omega_s}$

# List of symbols and notations

- $J_{tur}$  and  $J_{DFIG}$  : Aggregated inertias of the turbines and the generators
- $\tau_{em}$  and  $F$  : Electromagnetic torque of DFIG and coefficient of friction
- $P_m$  : Mechanical power
- $\tau_m$  : Mechanical torque
- $\omega_r$  : Rotor angular speed
- $P_s$  : Stator power
- $P_r$  : Rotor power
- $P_t$  : Total power

# Calculation for stator and rotor powers (actual values)

Wind speed = 9 m/s (constant)

Synchronous speed = 1200 RPM (126 rad/sec)

For a rated power of 1.5 MW, we calculate the stator and rotor contributions.

**Sub-Synchronous Mode** ( $s = 0.3016$ ,  $\omega_r = 88$  rad/sec)

$$P_{\text{stator}} = (1 - 0.3016) \times 1.5 \text{ MW} = 0.6984 \times 1.5 = 1.0476 \text{ MW}$$

$$P_{\text{rotor}} = 0.3016 \times 1.0476 = 0.316 \text{ MW}$$

- Stator supplies 1.0476 MW directly to the grid.
- Rotor draws 0.316 MW from the grid (converter absorbs power).

**Super-Synchronous Mode** ( $s = -0.2936$ ,  $\omega_r = 163$  rad/sec)

$$P_{\text{stator}} = (1 - 0.2936) \times 1.5 = 0.7064 \times 1.5 = 1.0596 \text{ MW}$$

$$P_{\text{rotor}} = -0.2936 \times 1.0596 = -0.311 \text{ MW}$$

# Summary of power flow

| Operating Mode    | Rotor Speed $\omega_r$ (rad/sec) | Slip $s$ | Stator Power (MW) | Rotor Power (MW)                 | Total Power (MW) | Converter Action      |
|-------------------|----------------------------------|----------|-------------------|----------------------------------|------------------|-----------------------|
| Sub-Synchronous   | 88                               | +0.3016  | 1.0476            | (-0.316)<br>(absorbed from grid) | 1.5              | Takes power from grid |
| Synchronous       | 126                              | 0        | 1.5               | 0                                | 1.5              | No power exchange     |
| Super-Synchronous | 163                              | -0.2936  | 1.0596            | (+0.311)<br>(delivered to grid)  | 1.5              | Injects power to grid |

- DC-link voltage

For a 575 V, 100 MVA, 60 Hz system

$$\left. \begin{aligned} V_{dc} &\geq \left( \frac{2\sqrt{2}}{\sqrt{3} * m} \right) V_r^{\max} \\ &= \frac{2\sqrt{2} * 575}{\sqrt{3} * 1} \\ &= 938.97 \text{ Volt} \\ \text{Safety factor} &= 22.5\% \\ &= 1.225 * 938.97 \\ &= 1150 \text{ Volt} \end{aligned} \right\}$$

- **RSC rating**

DFIG draws a lagging reactive power of 0.712 MVAr during motoring mode

$$\left. \begin{aligned} Q_r^* &= s \times 0.712 \text{ MVAr} \\ &= 0.214 \text{ MVAr} \end{aligned} \right\} \text{Eqn. 16}$$

$$\left. \begin{aligned} VA_{RSC} &= \sqrt{(P_r^*)^2 + (Q_r^*)^2} \\ &= \sqrt{0.3162^2 + 0.214^2} = 0.382224 \text{ MVA} \end{aligned} \right\} \text{Eqn. 17}$$

$$I_{RSC}^{rms} = \frac{VA_{RSC}}{\sqrt{3} \times V_r^{\max}} = \frac{382224.2}{\sqrt{3} \times 575} = 383.78 \text{ A} \quad \text{Eqn. 18}$$

$$I_{RSC}^{pk} = \sqrt{2} \times I_{RSC}^{rms} = \sqrt{2} \times 383.721 \text{ A} = 542.75 \text{ A} \quad \text{Eqn. 19}$$

$$I_{RSC}^{pp} = 7\% \text{ of } 542.75 \text{ A} = 38.002 \text{ A} \quad \text{Eqn. 20}$$

$$\begin{aligned} I_{RSC}^{IGBT} &= 1.3 \times \left\{ I_{RSC}^{pk} + I_{RSC}^{pp} \right\} \\ &= 754.98 \text{ A} \end{aligned} \quad \text{Eqn. 21}$$

The voltage rating is selected as 1150 V, AMM800GA176D is a suitable module available in the market

## • GSC Rating

$$I_{GSC}^{rms} = \frac{VA_{GSC}}{\sqrt{3} \times V_{xfmr}^{LV}} = \frac{1.154 \times 10^6}{\sqrt{3} \times 575} = 1158.5161 \text{ A} \quad \text{Eqn. 21}$$

$$I_{GSC}^{pk} = \sqrt{2} \times I_{GSC}^{rms} = \sqrt{2} \times 1158.5161 \text{ A} = 1638.3892 \text{ A} \quad \text{Eqn. 22}$$

$$I_{GSC}^{pp} = 7\% \text{ of } 1638.3892 \text{ A} = 114.6872 \text{ A} \quad \text{Eqn. 22}$$

$$I_{GSC}^{IGBT} = 1.3 \times \left\{ I_{GSC}^{pk} + I_{GSC}^{pp} \right\} \quad \text{Eqn. 23}$$
$$= 2278.93 \text{ A}$$

The voltage rating is selected similarly to RSC, as 1800 V, MG7405WZ is a suitable module available in the market

# Specifications of the system under study

|                               |                                        |                                                    |
|-------------------------------|----------------------------------------|----------------------------------------------------|
| Three-phase mutual inductance | Positive-sequence parameters           | $R_1 = 5.1842 \Omega$<br>$L_1 = 0.34379 \text{ H}$ |
|                               | Negative-sequence parameters           | $R_0 = 15.553 \Omega$<br>$L_0 = 1.0314 \text{ H}$  |
| No. of wind turbines          | 66                                     |                                                    |
| Frequency                     | 60 Hz                                  |                                                    |
| Pole pairs                    | 3                                      |                                                    |
| Output Voltage level of DFIG  | 575 V                                  |                                                    |
| Grid voltage level            | 161 kV                                 |                                                    |
| Transformer                   | 161 kV/575 V (Y/ $\Delta$ ), 115.5 MVA |                                                    |
| DC link voltage               | 1150 V                                 |                                                    |
| Rating of each wind turbine   | 1.5 MW                                 |                                                    |
| Grid voltage level            | 161 kV                                 |                                                    |
| Power rating                  | 100 MVA                                |                                                    |

|                     |          |
|---------------------|----------|
| Power coefficient   | 0.3755   |
| Tip speed ratio     | 6        |
| Gear ratio          | 1:166    |
| Wind turbine radius | 54.735 m |
| DC link voltage     | 1150 V   |
| Wind Speed          | 9 m/s    |

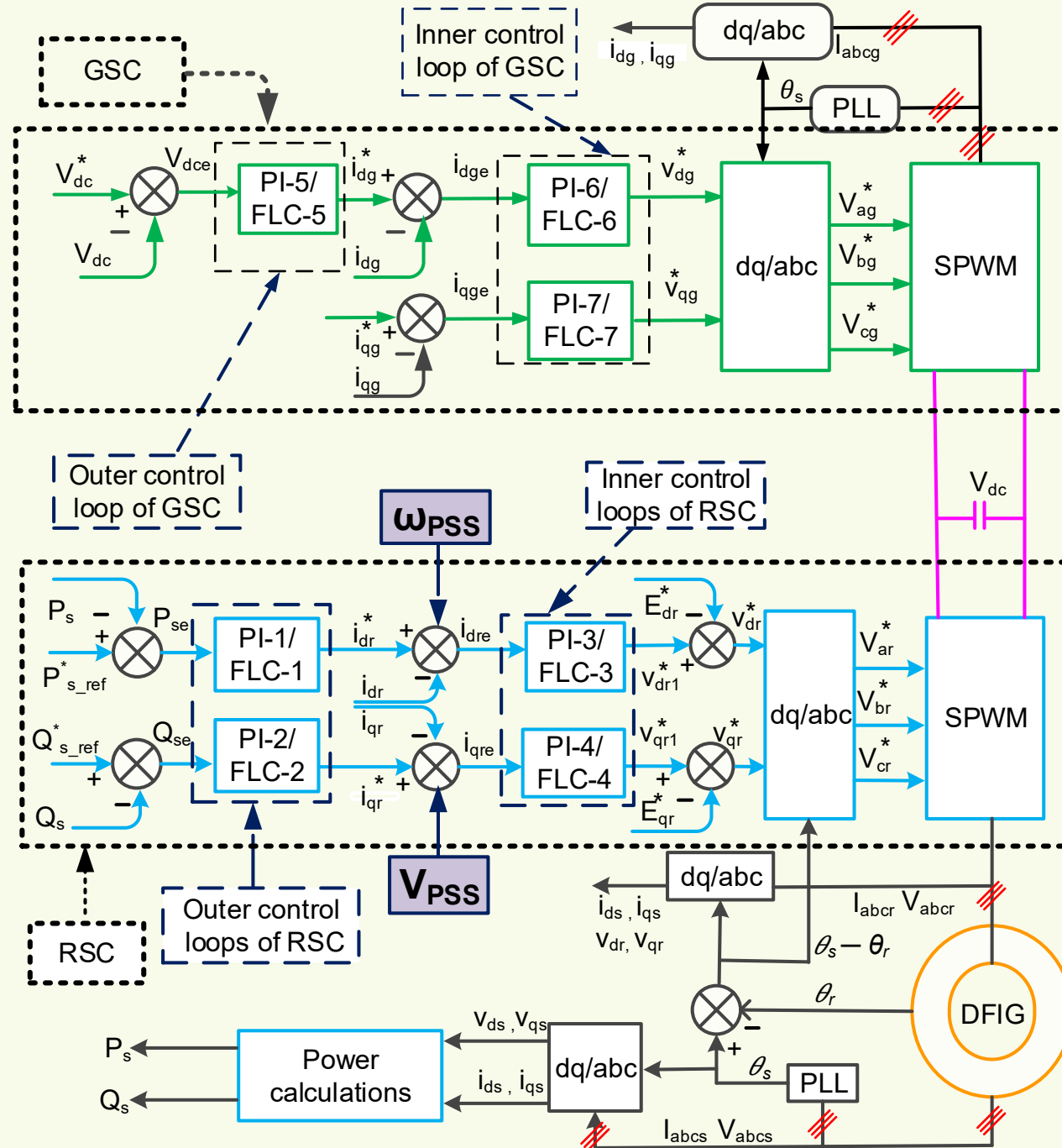
- Compensation level

$$\left. \begin{aligned} \%C &= \frac{X_C}{X_L} \times 100\% \\ X_C &= \frac{\%C}{100} \times X_L \end{aligned} \right\}$$

Eqn. 24

$$\left. \begin{aligned} X_C &= \frac{1}{2\pi fC} \\ C &= \frac{1}{2\pi f \left( \frac{\%C}{100} \right) \frac{V^2}{P}} \end{aligned} \right\}$$

Eqn. 25



SFOCS based PI/FLC for DFIG along with PSS controller to the GSC

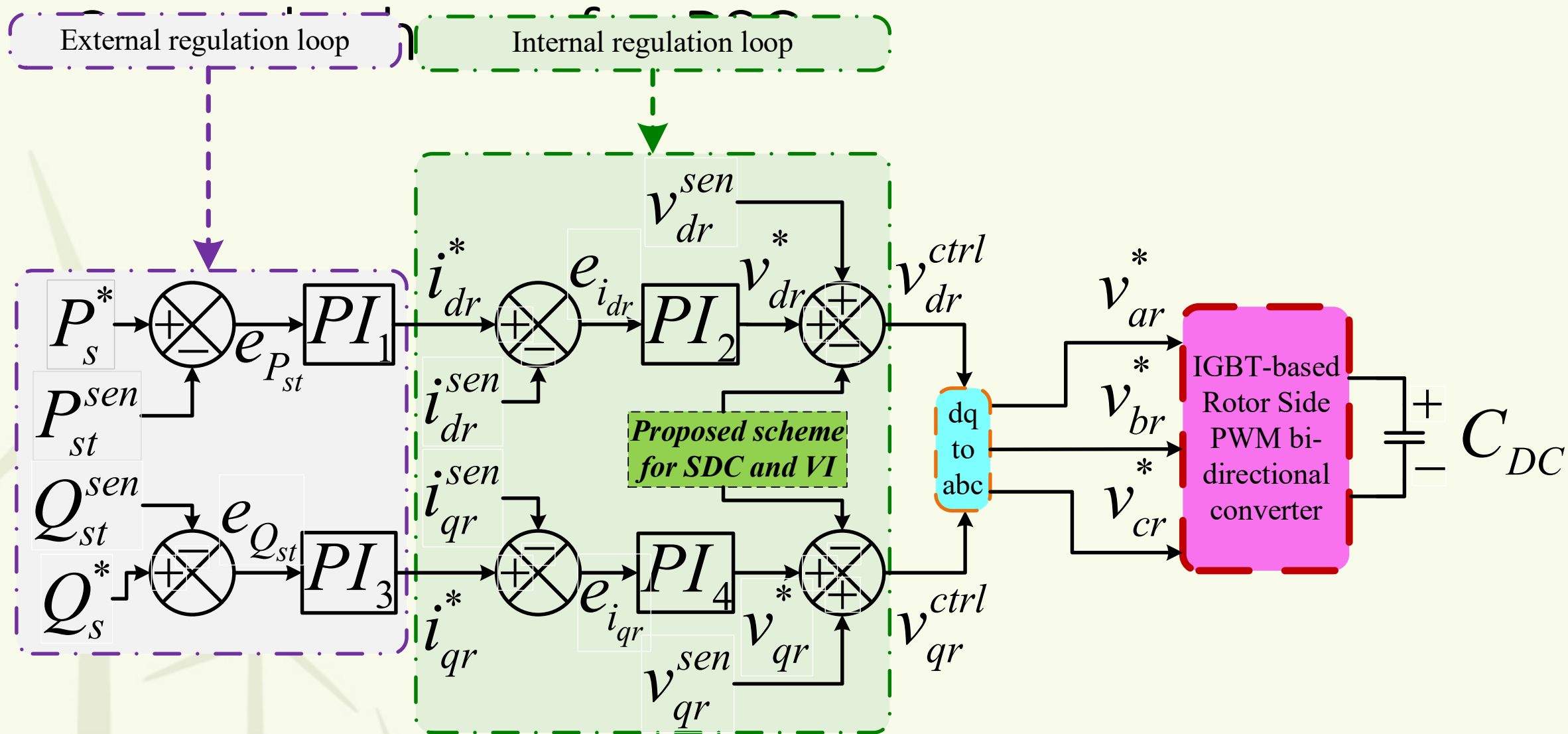


Fig. Control scheme: RSC

# Control scheme for GSC

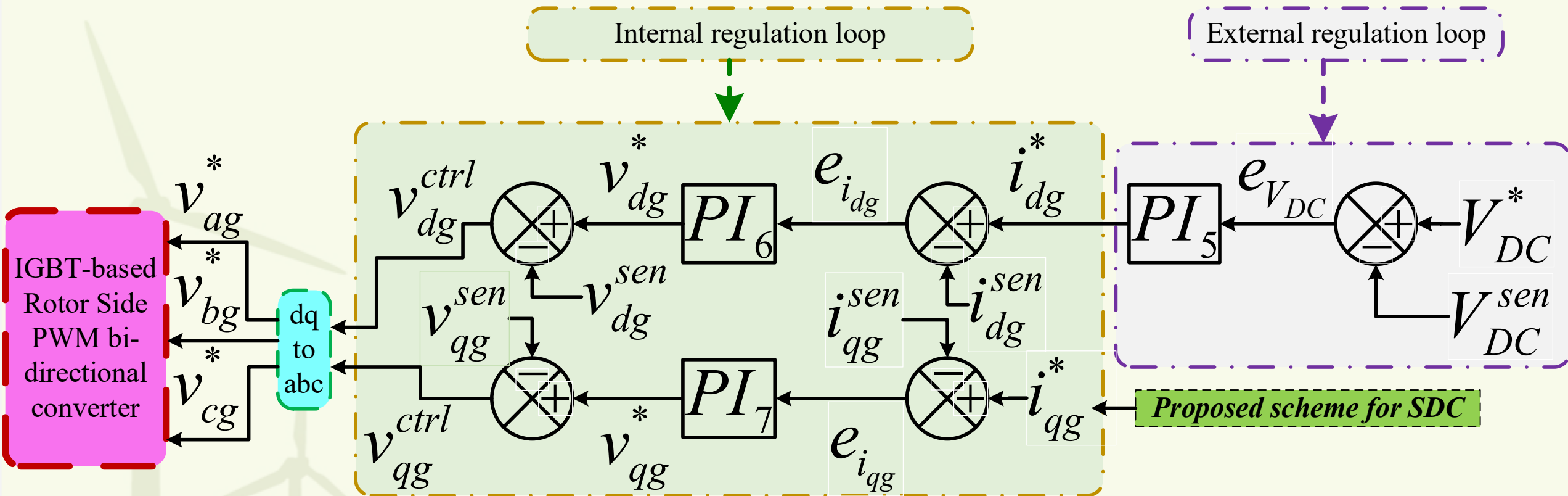


Fig. Control scheme: GSC

# Proposed Control Scheme for RSC

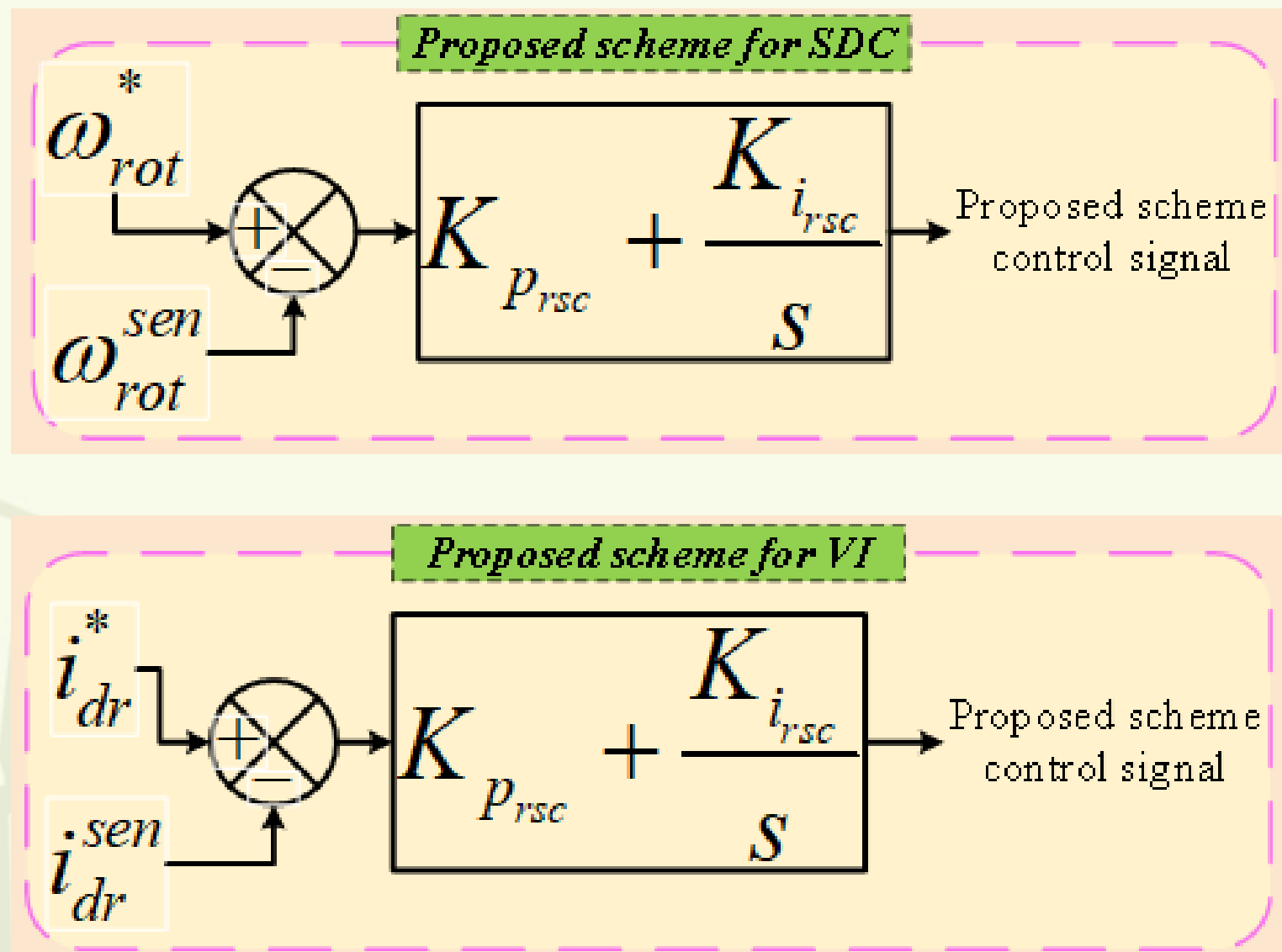


Fig. Proposed control scheme: RSC

# Proposed Control Scheme for GSC

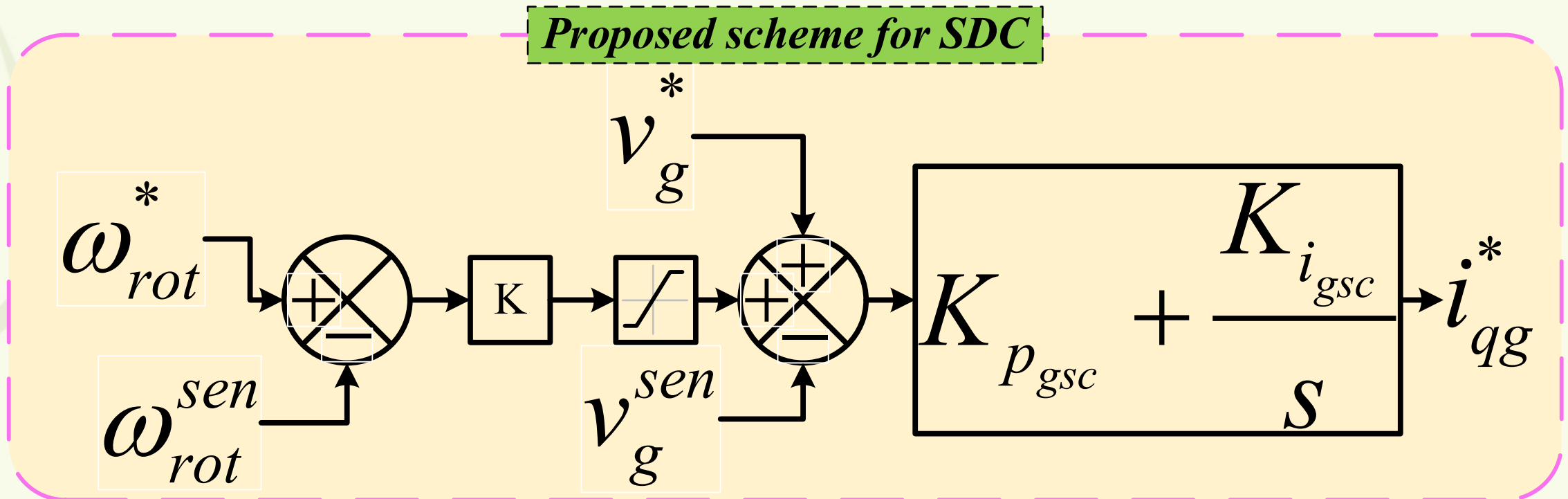


Fig. Proposed control scheme: GSC

# Modelling of SSR

- 3 levels of compensation

- 0 to 2 sec: 2%
- 2 to 4 sec: 20%
- 4 to 10 sec: 60%

- Cases of simulation

1. Without any controller
2. Controller only at GSC
3. Controllers at GSC + RSC
4. VI-SDC in RSC

- LLL-G fault

- Created at 2.5 sec
- Cleared at 2.6 sec

# Results

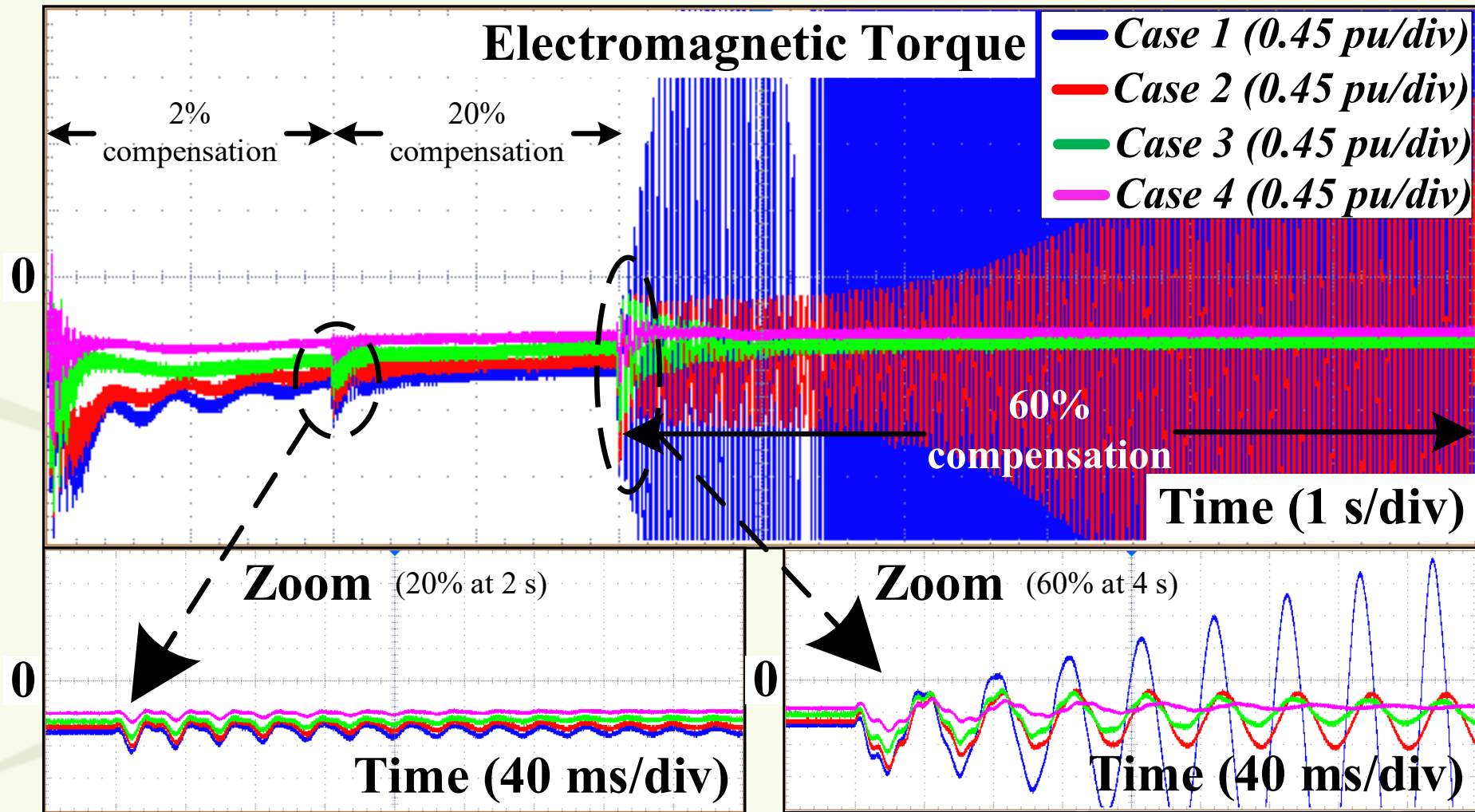


Fig. Electromagnetic torque subject to varying compensation levels

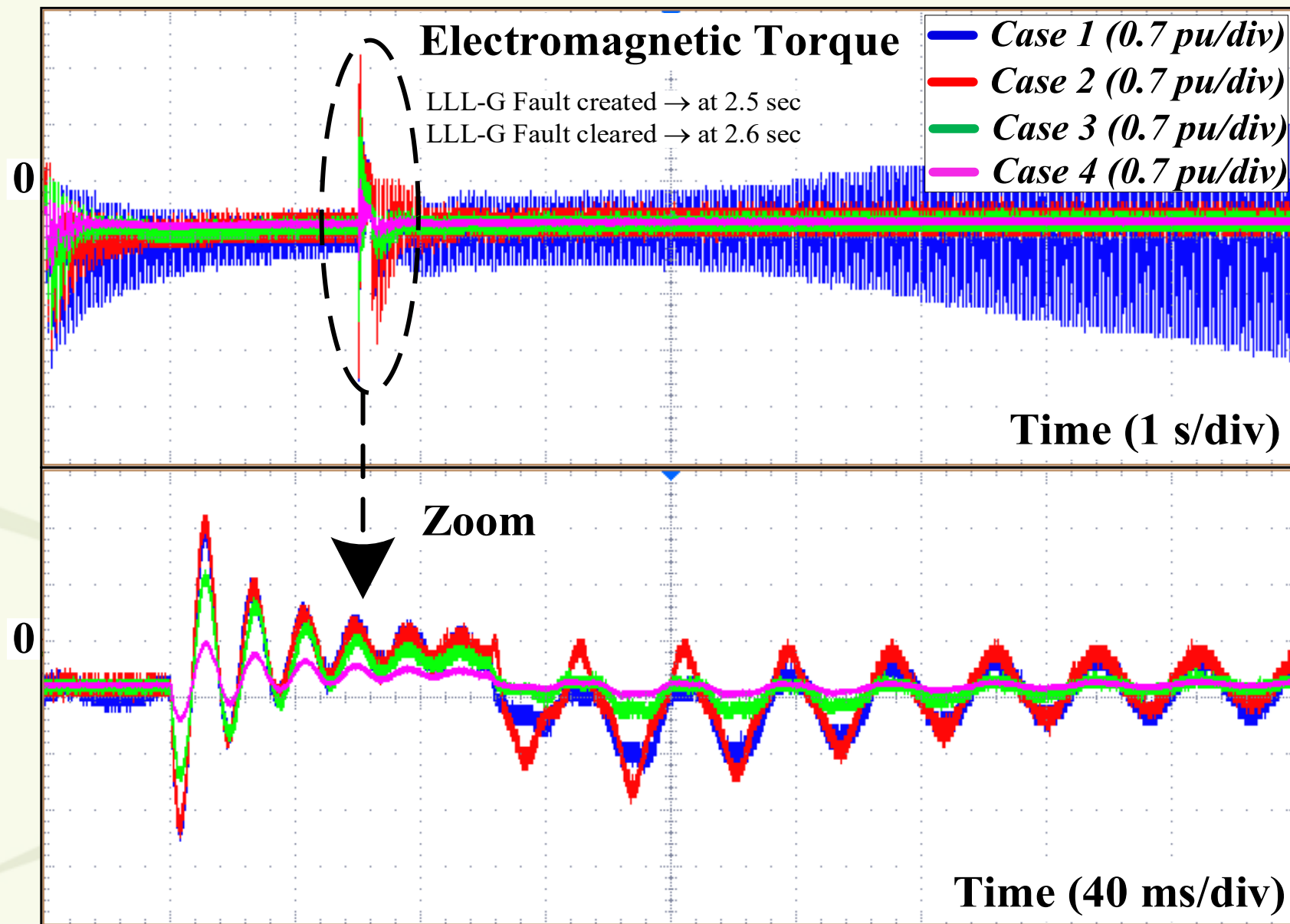


Fig. Electromagnetic torque subject to LLLG fault

# ■ Machine Learning application to DFIG based WTGs

- Objectives
- Introduction
- Series compensation and historical SSO event
- Conventional Methods to identify SSO modes
- Limitations of Conventional SSO Methods
- Data Driven Methods for SSO Studies
- Various ML classification algorithms used for SSO identification
- Simulation Results and Discussion
- Conclusion
- References

# Objectives



- 1) To design the DFIG-based wind farm based modified IEEE benchmark model
- 2) To apply the control strategy for the early prediction of sub-synchronous oscillation (SSO) conditions.
- 3) To study the different machine learning models for fast severity predictions
- 4) To collect the dataset and train the multilayer perceptron (MLP) model
- 5) Comparative analysis of MLP, SVM, LR, KNN, NB, XGB, AdaBoost, RF, and DT methods with training and testing accuracy
- 6) To test the performance evaluation of the model by using the confusion matrix

# Why a need for series compensation & Advantages of series compensation



- A new wind energy power plants can be integrated into the existing national grid through dedicated or existing transmission lines by increasing the power transfer capability using series compensation (SC) or FACT devices [8]. Installing SC in a long transmission line is an inexpensive and efficient way to improve the capability of power transfer of an existing transmission line [9].

## ❖ Why did SSR occur in WF?

- The problem that may arise because of SC in a long transmission line is Sub-synchronous resonance (SSR), which can lead to shaft failure and power system instability [10],[11]. Also, the expansion of renewable generation significantly increases the severity and possibility of sub-synchronous oscillation (SSO) issues in modern power systems.

**Table 1- Major SSO Events**

| Year | Location     | SSO Type  | Severity | Reference          |
|------|--------------|-----------|----------|--------------------|
| 2015 | Hami         | SSCI      | Failure  | [3], [4], [5]      |
| 2012 | Guyuan       | SSR, SSCI | Failure  | [6], [7]           |
| 2011 | Shangdu      | SSR       | Test     | [4], [6], [10]     |
| 2009 | ERCOT        | SSR, SSCI | Failure  | [9], [1], [4]      |
| 1987 | Chester      | SSTI      | Test     | [8], [2], [5], [6] |
| 1983 | Ontario      | SSTI      | Test     | [4]                |
| 1980 | Square butte | SSTI      | Test     | [7], [9], [8], [4] |
| 1978 | Cholla       | SSR       | Test     | [7], [5], [1]      |
| 1976 | Navajo       | SSR       | Test     | [5], [2], [3]      |
| 1970 | Mahave       | SSR       | Failure  | [2], [3], [1]      |

- Sub-synchronous oscillations (SSO) in wind farms occur for two main reasons: (1) interaction of wind turbines with the series compensated grid and (2) interaction of wind turbines with the weak grid [16]. The severity of SSO depends on several factors, among which wind speed, level of series compensation, and number of wind turbines in service can be identified as major contributors.
- Accordingly, lower wind speeds, higher compensation levels, or an increasing number of wind turbines will worsen the severity of SSO. However, some SSOs occur even at low levels of series compensation. The 2012 Guyuan incident can be cited as an example.

### ❖ **Solution of SSR Occurrence.**

- Therefore, to mitigate the impact of SSOs, early prediction of their dominant severity modes is essential for developing effective control strategies. This research proposes to develop an algorithm to anticipate severe SSO and send early warning signals to activate necessary mitigation controls, thereby preventing the escalation of these events.

# Conventional Methods vs Machine Learning in Power Systems



| Feature/Aspect            | Conventional Methods             | Machine Learning                                   |
|---------------------------|----------------------------------|----------------------------------------------------|
| Modeling Approach         | Equation-based (deterministic)   | Data-driven (probabilistic or pattern-based)       |
| System Complexity         | Limited to simple/linear systems | Handles nonlinear and complex systems              |
| Adaptability              | Low — requires manual tuning     | High — adapts automatically with new data          |
| Data Usage                | Limited use of data              | Learns directly from historical and real-time data |
| Fault Detection           | Reactive (after-the-fact)        | Predictive (early detection)                       |
| Load Forecasting Accuracy | Lower for irregular patterns     | Higher, especially with variable loads             |
| Scalability               | Hard to scale with system growth | Scales easily with more data & variables           |
| Pattern Recognition       | Poor unless explicitly modeled   | Excellent — finds hidden trends automatically      |
| Handling Uncertainty      | Sensitive, needs assumptions     | Robust to noise and uncertainty                    |
| Maintenance Optimization  | Manual or rule-based scheduling  | Predictive maintenance and automated optimization  |



Machine learning (ML) methods have been employed for mode identification in SSO events [27].

- 1) These methods effectively extract nonlinear electrical characteristics
- 2) It can quickly deliver mode predictions based on online measurement data.

**Table 3- Previous studies literature summary of machine learning methods to solve SSO problems**

| ML Classifier                                    |                                                                                                                                                                                                                                                                                                                      |      |
|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| Multiple support vector machine (SVM) classifier | In this method, three features reflecting different SSO and interference levels are identified by analyzing historical PMU data and used as inputs to the multi-SVM classifier.                                                                                                                                      | [28] |
| Convolutional neural network (CNN)-based method  | For mode identification, utilizing a comprehensive SSO dataset                                                                                                                                                                                                                                                       | [29] |
| Long Short-Term Memory (LSTM)                    | Neural Network approach for estimating the dominant mode based on Synchrophasor data                                                                                                                                                                                                                                 | [30] |
| Extreme Learning Machine (ELM)--based classifier | For the rapid and accurate recognition of SSOs                                                                                                                                                                                                                                                                       | [31] |
| SVM algorithm                                    | The Fast Fourier Transform (FFT) is used to extract the sub-synchronous frequency components from the PMU current phasor. used to find harmful SSO components. Three factors, namely, amplitude attenuation, frequency susceptibility, and energy accumulation, are then used to determine whether to send an alarm. | [32] |
| Decision tree (DT)                               | For suspected SSO, FFT analysis is performed to calculate the amplitudes and frequencies of SSO components. Then, an alarm signal will be sent if the frequency is close to the torsional vibration frequency of the generator shaft, the amplitude is high, or it lasts for extended periods.                       | [33] |

# System Under Study

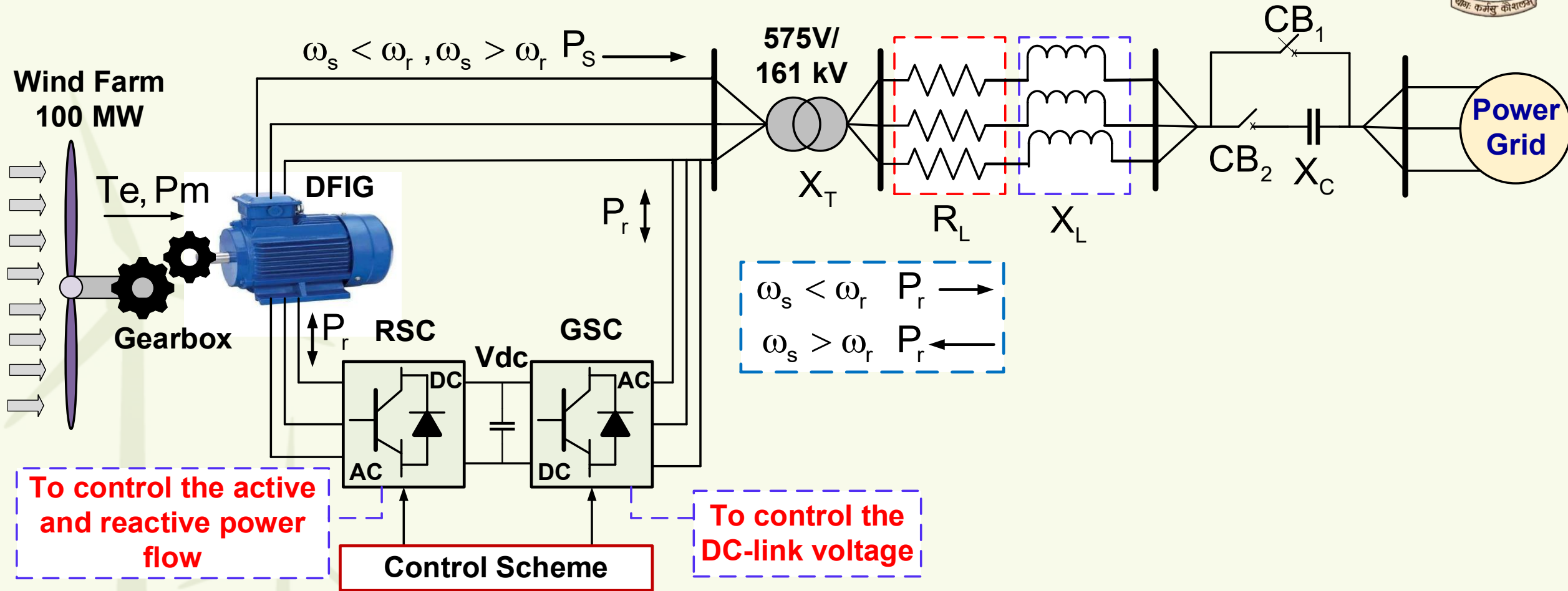


Fig. 1. Modified IEEE First Benchmark Model

- When a small WF, whenever an SSO occurred, that time, a PS engineer can easily tell whether the system is non-SSO (stable), Non-severe SSO, or Severe SSO, but when have a large WF, it is not possible. Suppose 1000 connecting DFIGs, then that particular case PS engineer can't say system is Non-SSO (stable), Non-severe SSO or Severe SSO has occurred, in which individual DFIG generators' by using graphical analysis. It is very critical to identify the system's behavior. So, to overcome these issues, the proposed research is focused on ML algorithms that can be easily applied to solve the above problem.

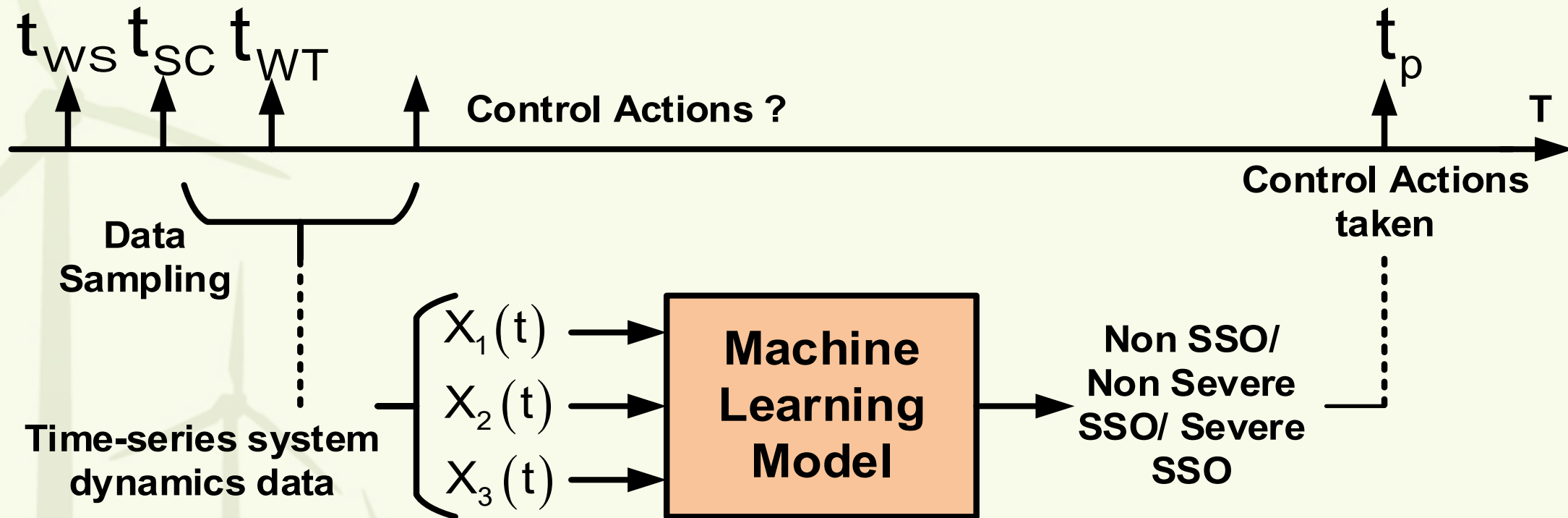
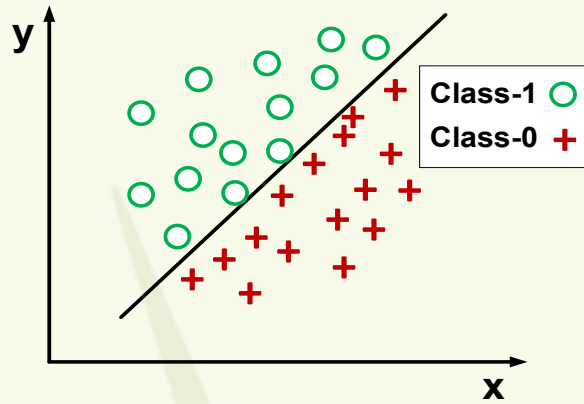
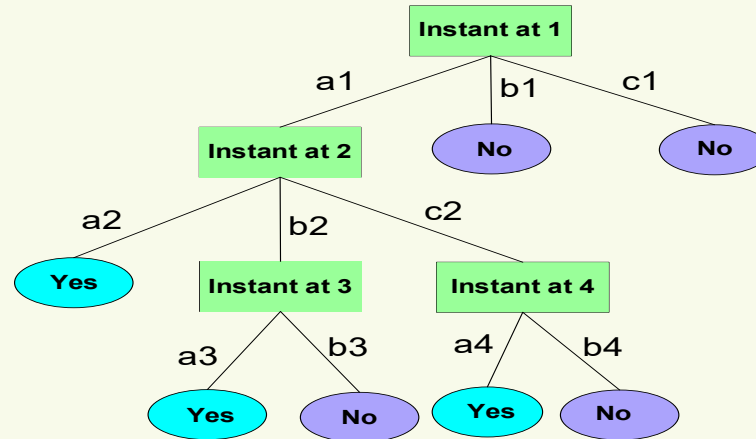


Fig. 2. Illustration of the severity prediction task

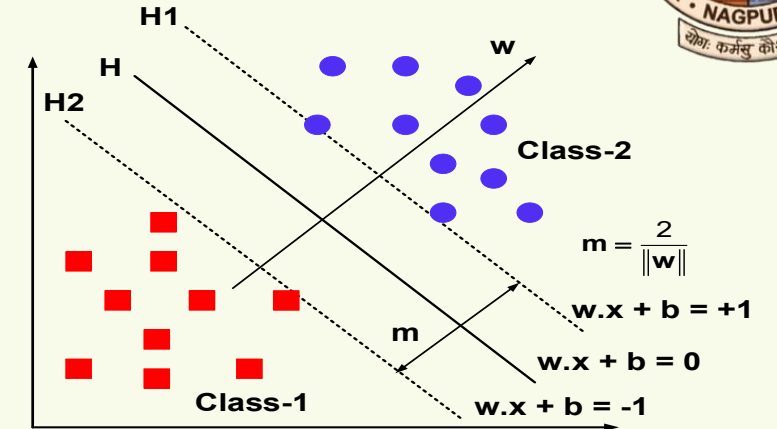
# Machine Learning Models for Fast Severity Predictions



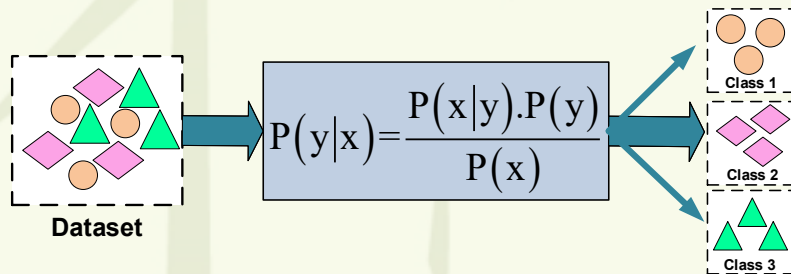
(a) Logistic Regression (LR) algorithm



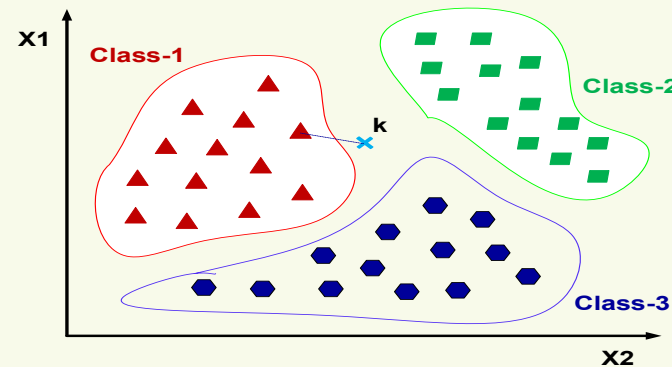
(b) Decision Tree (DT) algorithm



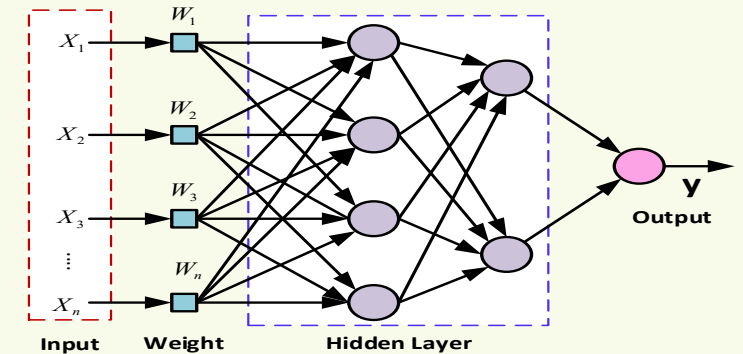
(c) Support Vector Machine (SVM) algorithm



(d) Naïve Bayes (NB) algorithm

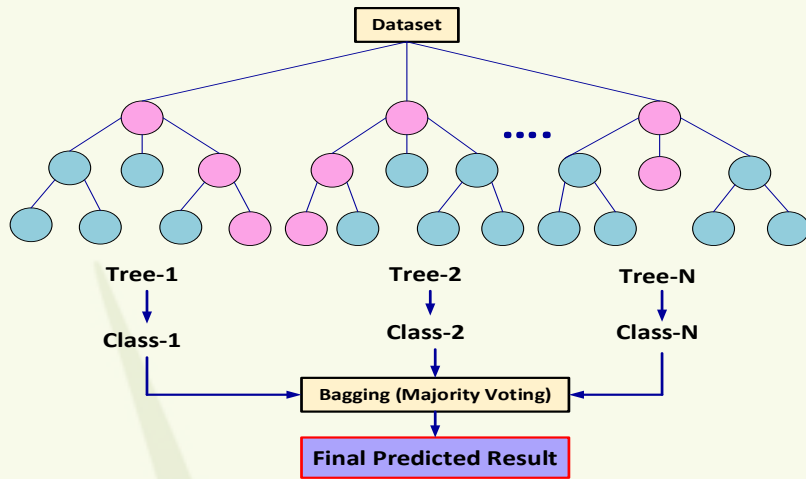


(e) K-Nearest Neighbour (KNN) algorithm

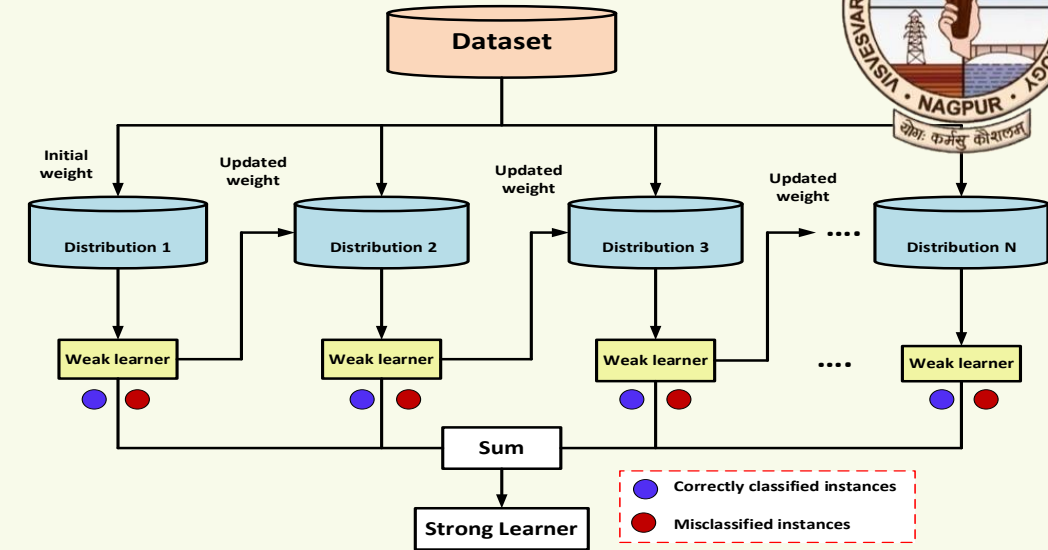


(f) Multilayer Perceptron (MLP) algorithm

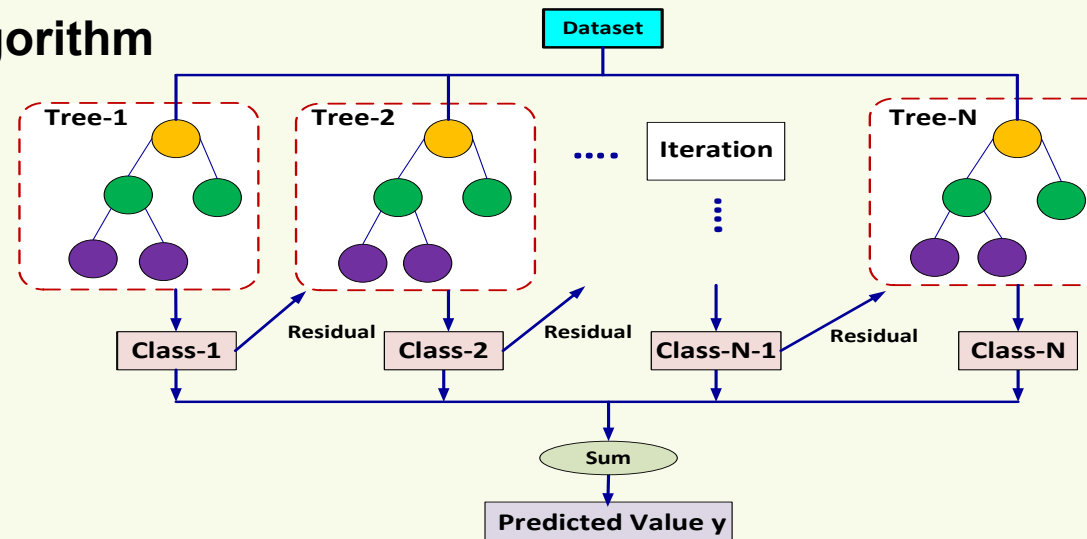
# Machine Learning Models for Fast Severity Predictions



(g) Random Forest (RF) algorithm



(h) Adaptive Boosting (AdaBoost) algorithm



(i) XGBoost (XGB) algorithm

Fig. 3. Classification algorithm for fast SSO prediction

# Developing the training dataset

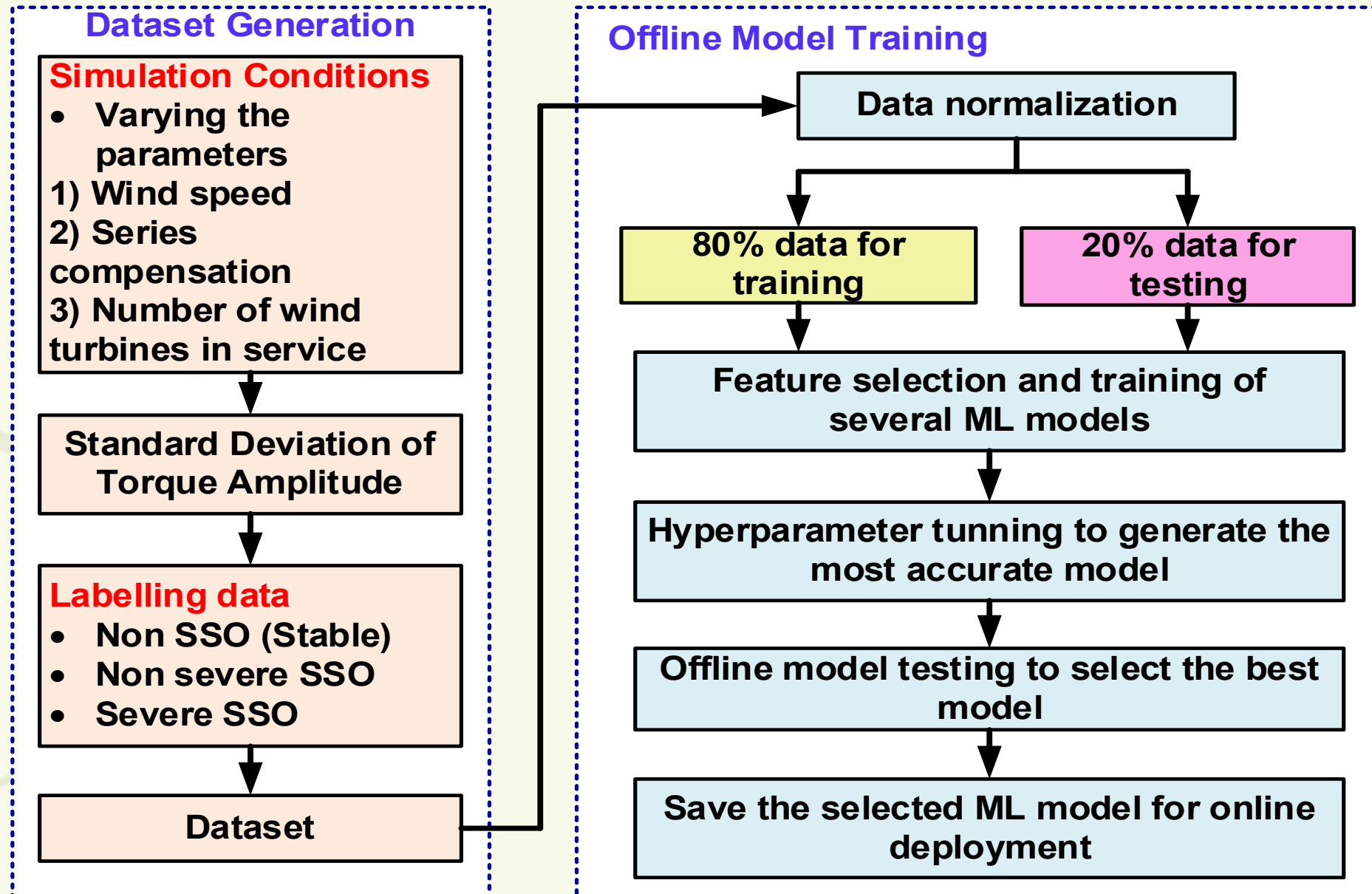


Fig. 4. Classification of TSA Study

# Training dataset labelling criteria



**Table 4- The summary and comparisons of previous reported studies to label the dataset**

| Ref.          | Label determination method                                                                                     | Class-1                                                                                                                                                                             | Class-2                                                                                                                                                   | Type classifiers     |
|---------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| [26]          | Using FFT spectral analysis                                                                                    | SSO Feature Vectors: These are labelled as "+1" to indicate the presence of sub-synchronous oscillations.                                                                           | Non-SSO Feature Vectors: These are labelled as "-1" to signify the absence of sub-synchronous oscillations                                                | Multi-SVM classifier |
| [30]          | According to the parameters K, M, and N established by employing the amplitude of percentage                   | If a sample's amplitude percentage exceeds a specific large value (denoted as M%), it is likely to be a real SSO component. Such samples are labelled as "need to pay attention"    | If a sample's amplitude percentage is below a certain low value (denoted as N%), it is considered non-harmful and labelled as "no need to pay attention." | SVM classifier       |
| [31]          | Components K and M, defined by amplitude and saliency, indicate the amount of frequency amplitude fluctuation. | A feature vector can be labelled as "Suspected SSO" if there is at least one sub-synchronous component whose amplitude percentage exceeds a predetermined threshold, denoted as K%. | A feature vector is derived from "Ambient" based on the characteristics of the data.                                                                      | DT classifier        |
| Proposed work | Torque Amplification (TA)                                                                                      | HTAS: If a sample's TA exceeds a specific large value, it is likely to be a real SSO component and such samples are labelled as "Severe SSO".                                       | LTAS: if a sample's TA is below a certain low value, it is considered non-harmful and labelled as "Non SSO."                                              | MLP classifier       |

# Multilayer Perceptron Model Training

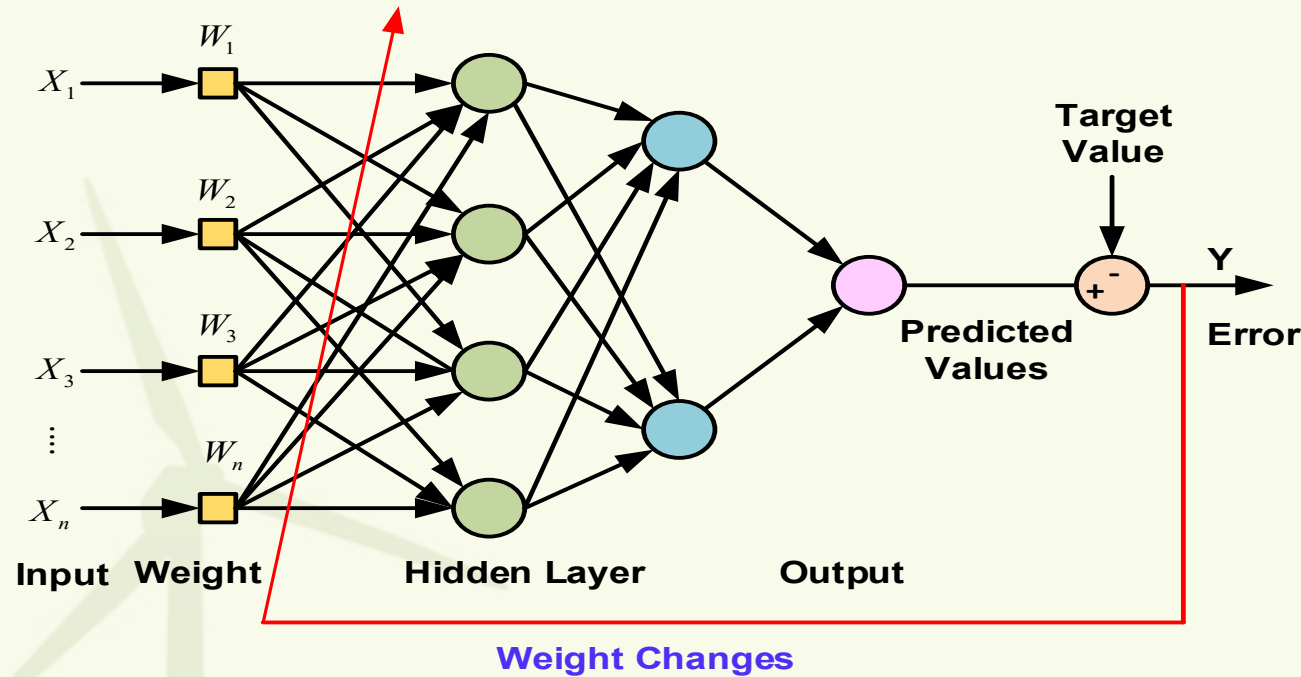


Fig. 5. MLP algorithm with updating weights

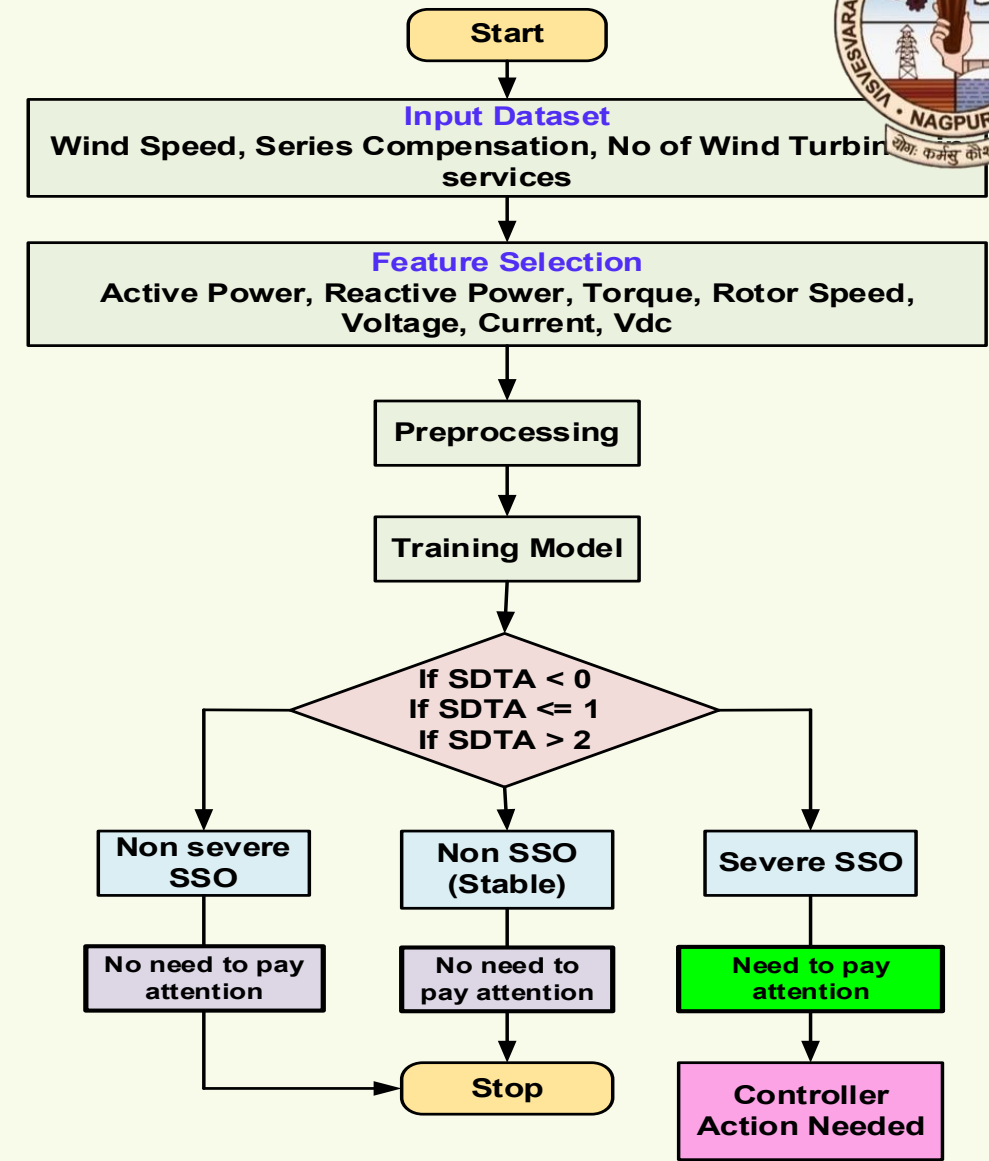


Fig. 6. Implementation of the MLP algorithm into the proposed work

- The study involved the development of a dataset comprising 10,649 operating conditions, achieved through variations in WS, SC, and WT.
- The WS below the cut-in WS (3 m/s) and cut-out WS (25 m/s). To generate data points for the level of SC, a range of 15% to 80% is used, while the WT varies between 31-67 (50%-100% of the capacity of the WF). These will be inputs to the simulation model in Figure 1.
- The simulation model initially operates with minimal SC (CB1 closed, CB2 opened). At  $t = 2\text{s}$ , it is connected to a high SC level to simulate the system's response when a WF is radially connected to a series-compensated line (CB1 opened, CB2 closed). It is assumed that the circuit breakers of the disconnected transmission line successfully re-close after a 1-second interval, returning the system to its initial operating configuration at  $t = 3\text{s}$  (CB1 closed, CB2 opened).

| Parameters                         | Range          | Variable Step |
|------------------------------------|----------------|---------------|
| Wind Speed                         | 3 m/s – 25 m/s | 1             |
| Series compensation                | 15 % - 90%     | 1             |
| Number of wind turbines in service | 31- 67         | 1             |

# Different SC level with SSO conditions

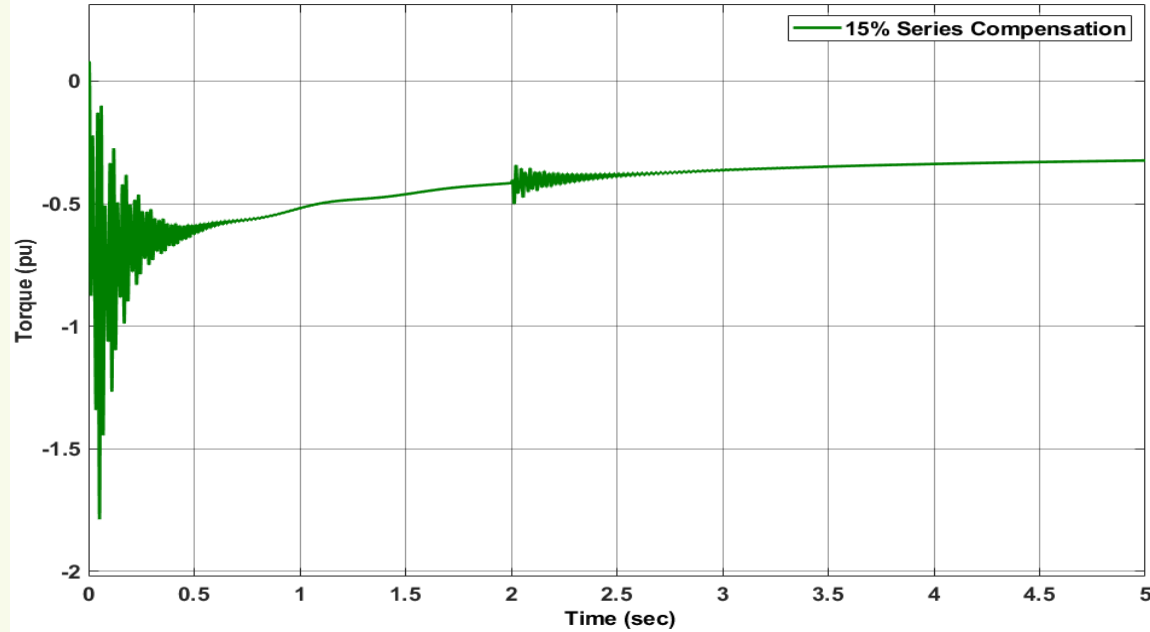


Fig. 7. Non-SSO (stable) condition

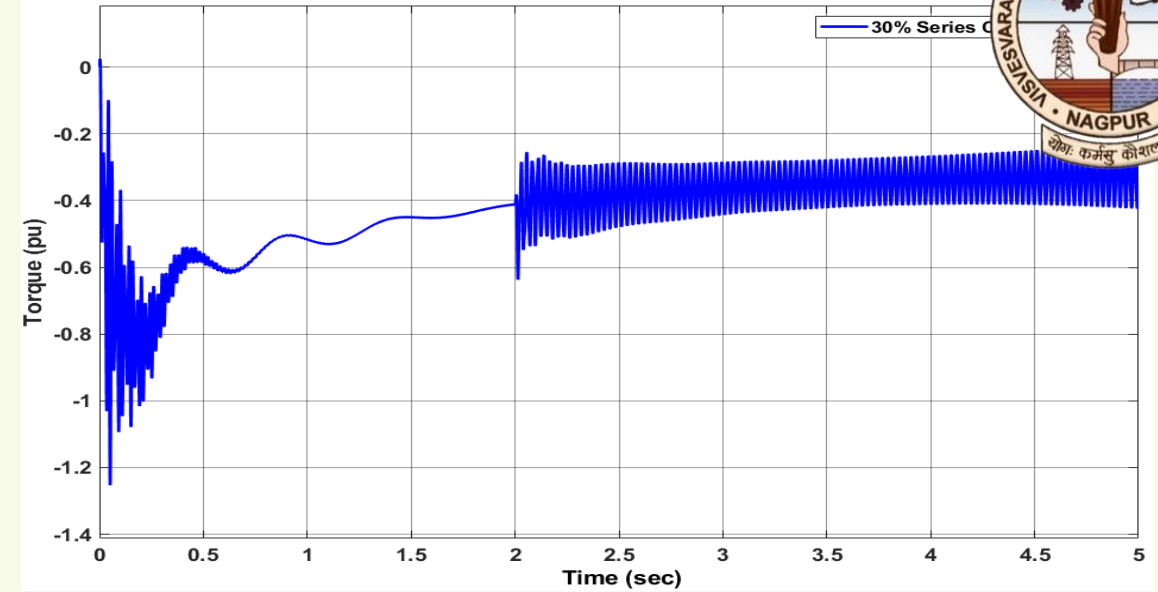


Fig. 8. Non-severe SSO condition

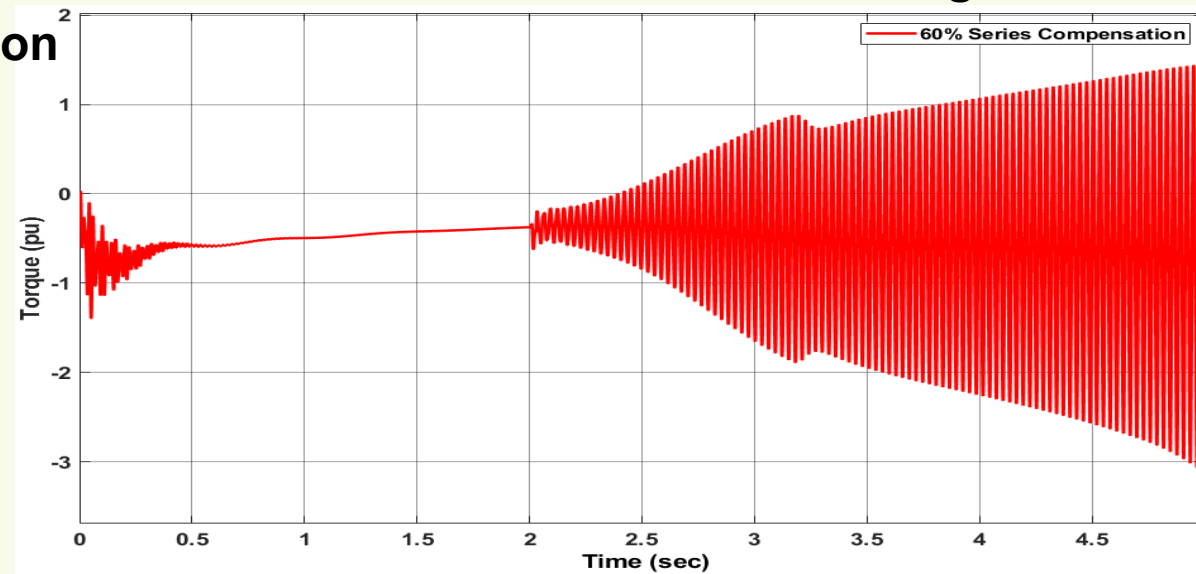
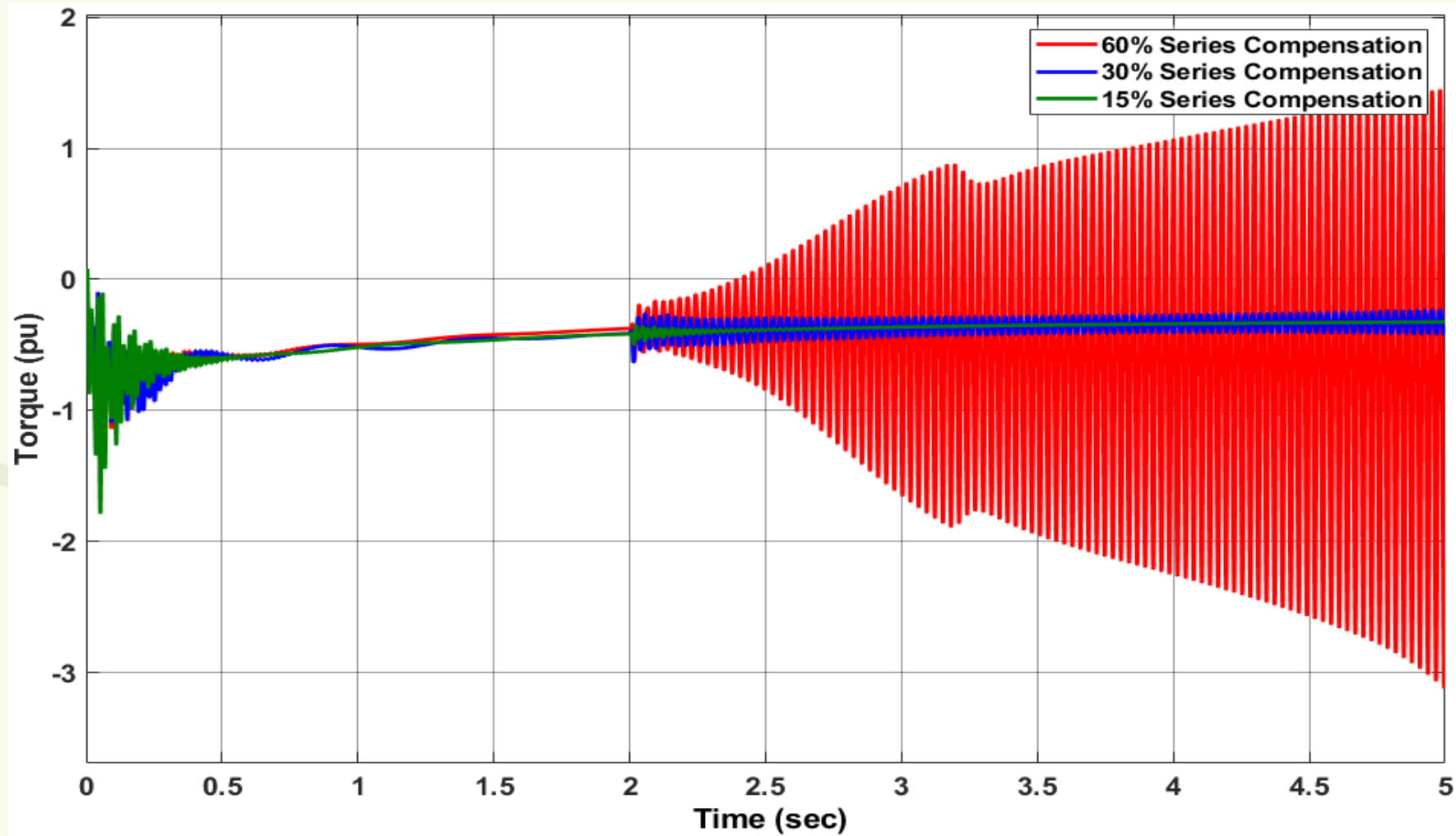


Fig. 9. Severe SSO



**Fig. 10. Dynamic behaviour of the electromagnetic torque ( $T_e$ ) of different series compensation levels at 7 m/s wind speed: an SSO conditions analysis for (a) 15% series compensation; (b) 30% series compensation; and (c) 60% series compensation**

# Distribution of input feature set- WS, SC, WT

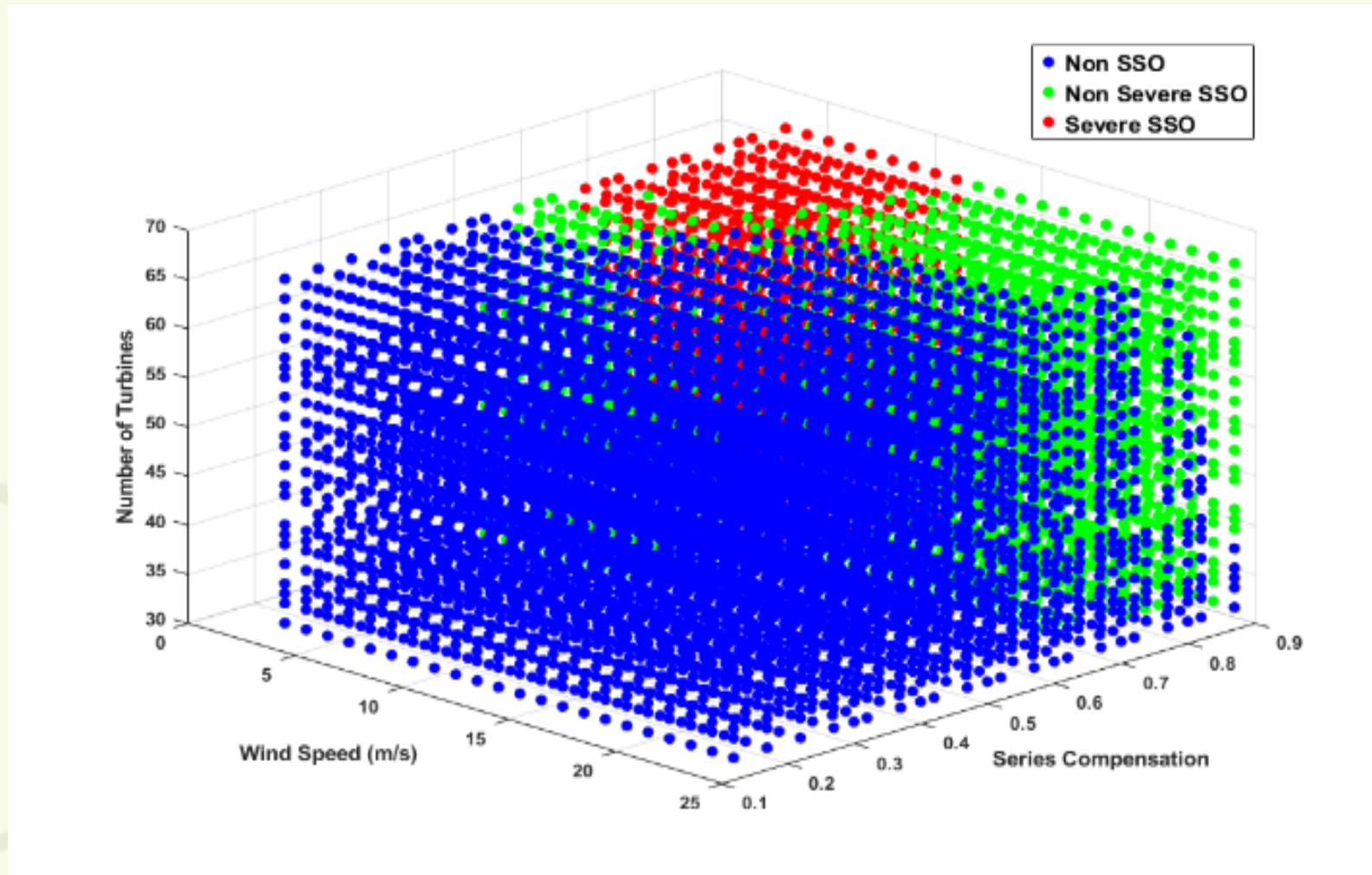
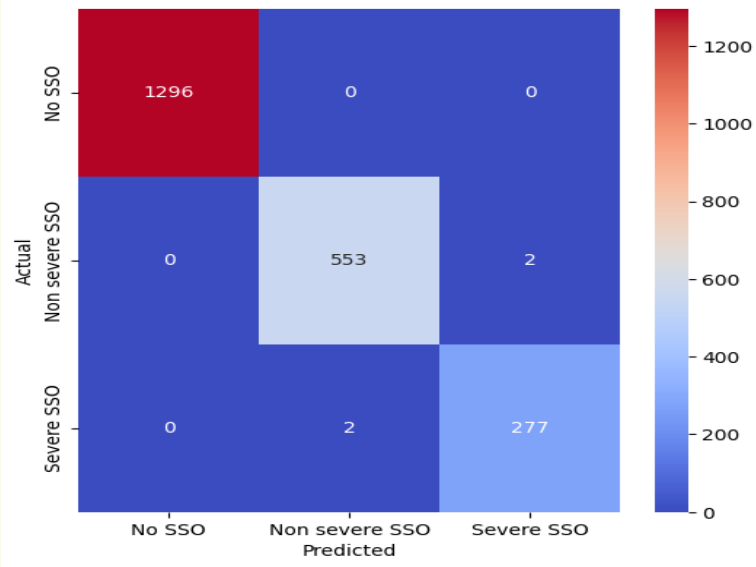
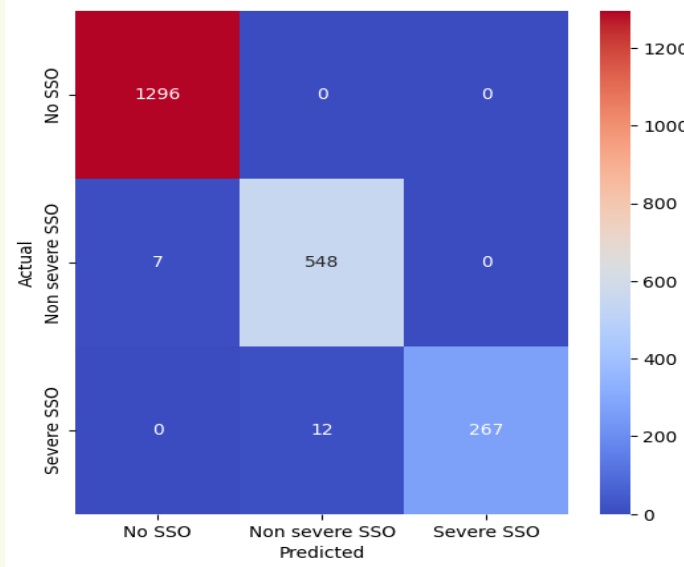


Fig. 11. Distribution of input feature set- WS, SC, WT

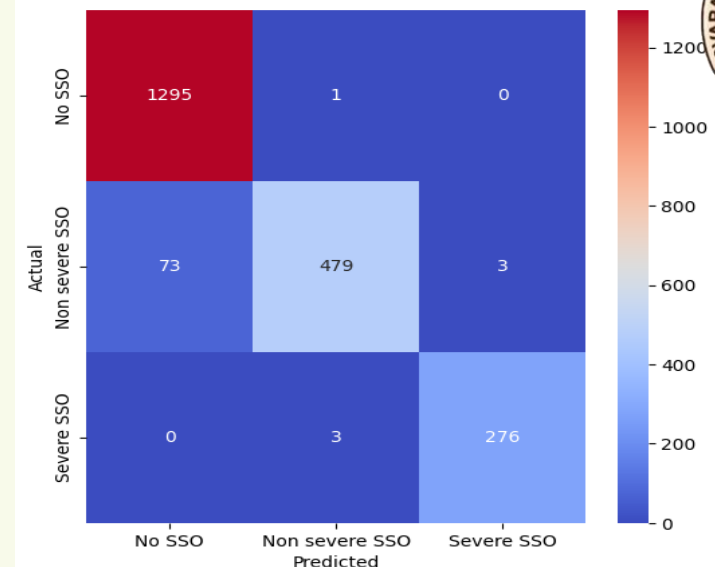
# Confusion Matrix



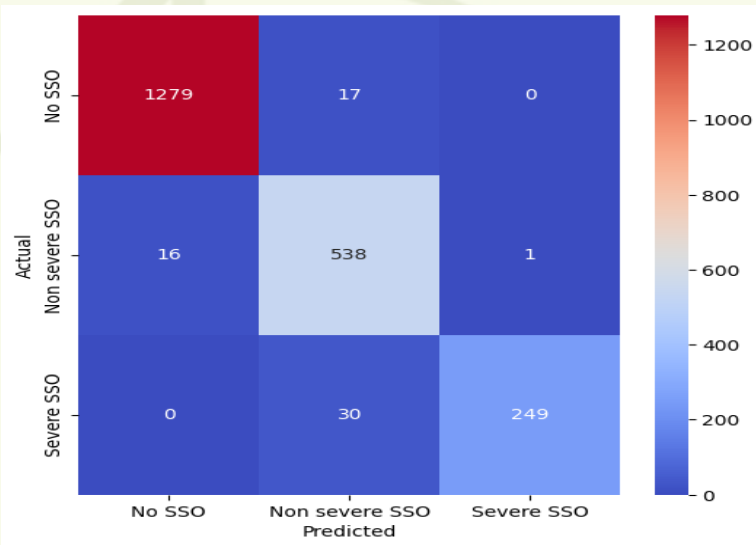
(a) DT Classifier (Accuracy: 99.81%)



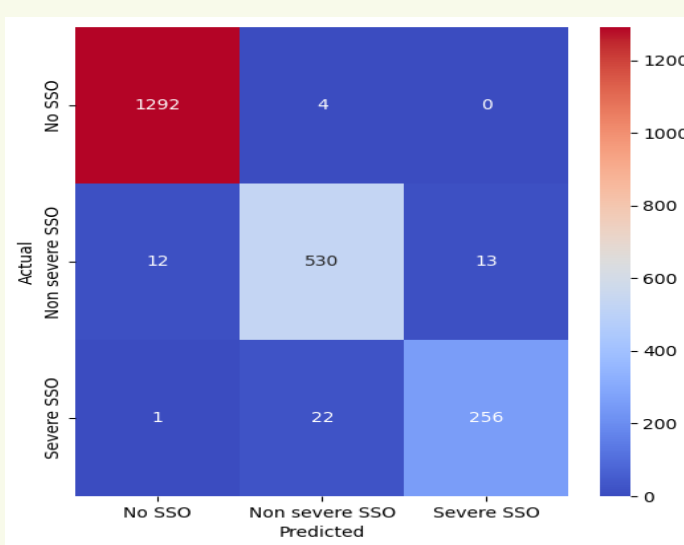
(b) MLP Classifier (Accuracy: 99.01%)



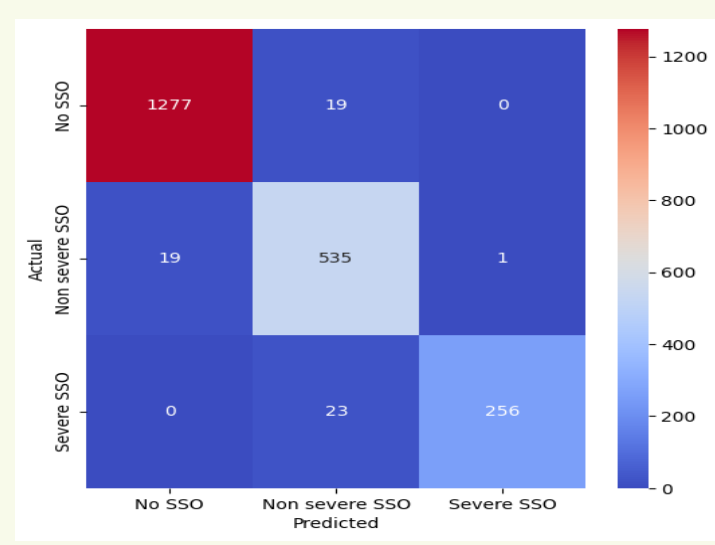
(c) NB Classifier (Accuracy: 96.24%)



(d) SVM Classifier (Accuracy: 97.00%)

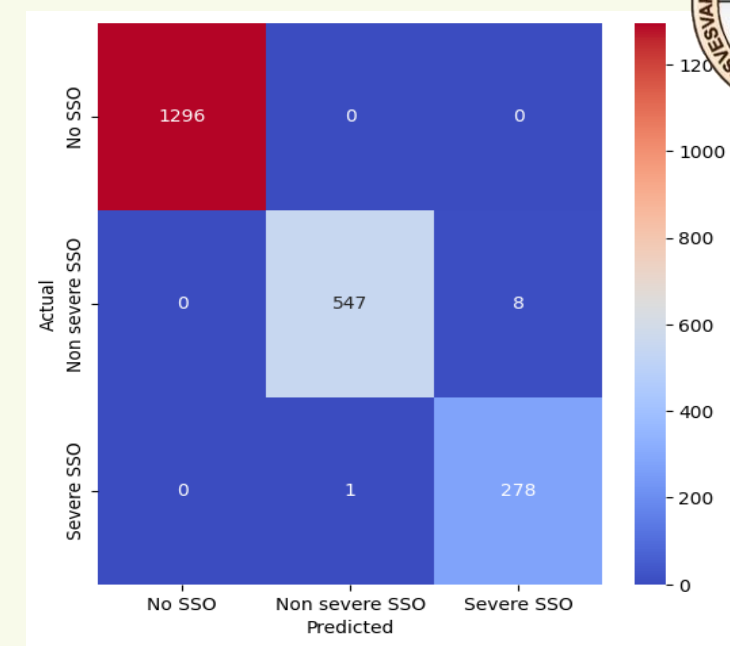
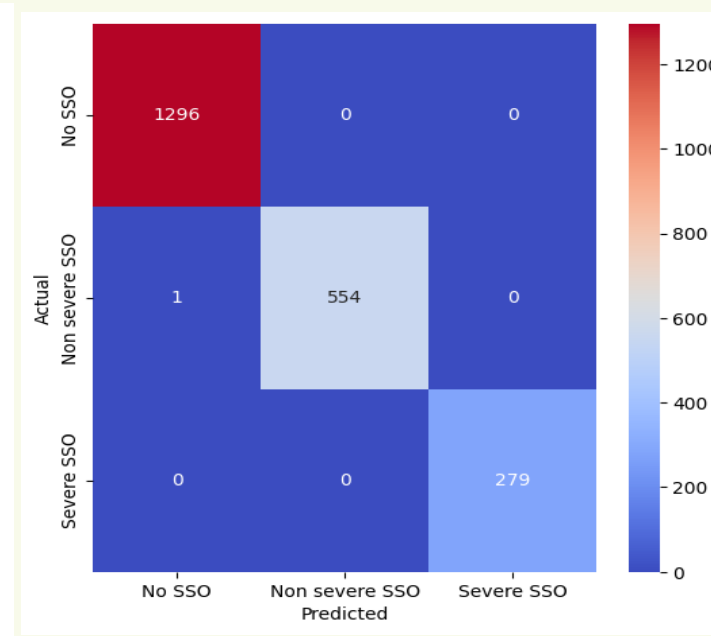
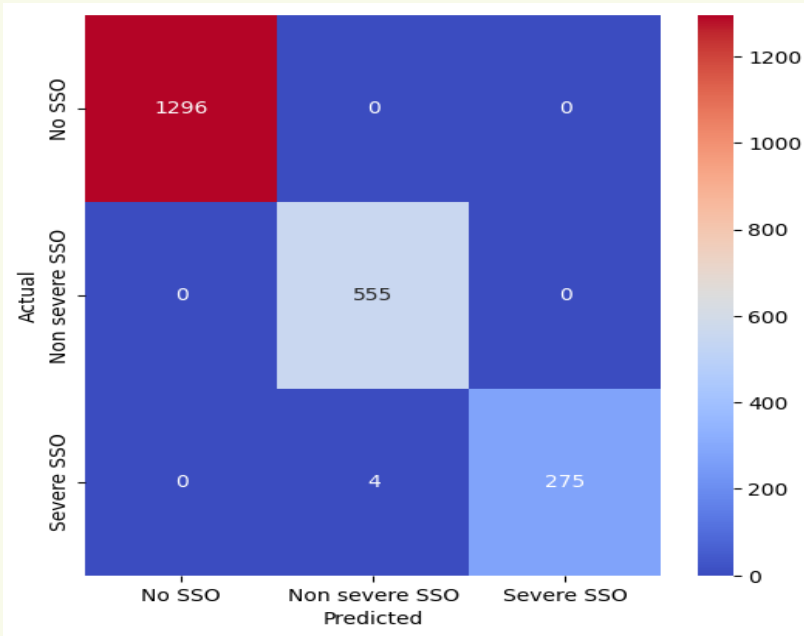


(e) KNN Classifier (Accuracy: 97.56%)



(f) LR Classifier (Accuracy: 97.09%)

# Confusion Matrix



(g) RF Classifier (Accuracy: 99.81%)

(h) XGB Classifier (Accuracy: 99.95%)

(i) AdaBoost Classifier (Accuracy: 99.58%)

**Fig. 12. Confusion matrix for test dataset of classification, i.e., concerning SSO event zone, obtained by various classifiers: (a) DT classifier, (b) MLP classifier, (c) NB classifier, (d) SVM classifier, (e) KNN classifier, (f) LR classifier, (g) RF Classifier, (h) XGB Classifier, (i) AdaBoost Classifier.**

**1) Accuracy:** In classification evaluation, using accuracy (ACC) is beneficial because it succinctly quantifies the proportion of correctly classified instances relative to the total number of instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

**2) Precision:** The precision (also called positively predictive value) describes the capability of a model to identify a severe SSO condition's overall true positive prediction. It is represented as the ratio of the correct positive predictions to the number of samples labeled as positive.

$$\text{Precision} = \frac{TP}{TP + FP}$$

**3) Recall:** The recall (also called sensitivity) gives the model the capability to identify all severe SSO conditions. Recall is described as the ratio of the number of correct positive predictions to the number of positive samples.

$$\text{Recall} = \frac{TP}{TP + FN}$$

**4) F1\_Score:** The F1-score is the harmonic mean of precision and recall, which inherently implies that a high F1-score can only be achieved if both precision and recall are high. If either is low, the F1-score will tend toward the lower value.

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

**Table 6- Results of sequential feature selection**

| Input features    | Accuracy (%) |       |       |       |       |       |       |       |          |
|-------------------|--------------|-------|-------|-------|-------|-------|-------|-------|----------|
|                   | NB           | DT    | MLP   | SVM   | KNN   | LR    | RF    | XGB   | AdaBoost |
| <b>WS, SC, WT</b> | 87.42        | 98.69 | 98.54 | 96.67 | 96.95 | 90.05 | 98.78 | 98.73 | 77.32    |
| <b>WS, SC</b>     | 87.09        | 96.15 | 94.84 | 95.63 | 96.06 | 90.14 | 96.15 | 96.15 | 78.26    |
| <b>WS, WT</b>     | 61.36        | 57.00 | 61.41 | 61.17 | 51.83 | 60.99 | 56.90 | 57.42 | 60.94    |
| <b>SC, WT</b>     | 75.87        | 77.79 | 78.03 | 78.03 | 72.63 | 76.48 | 77.09 | 77.79 | 76.20    |
| <b>WS</b>         | 61.08        | 60.42 | 60.85 | 60.85 | 54.51 | 60.99 | 60.42 | 60.42 | 60.42    |
| <b>SC</b>         | 76.95        | 78.03 | 78.03 | 78.03 | 72.96 | 76.95 | 78.03 | 78.03 | 78.03    |
| <b>WT</b>         | 60.85        | 60.85 | 60.85 | 55.92 | 60.85 | 60.85 | 60.85 | 60.85 | 60.85    |

**Table 7- Results of classification model performance in training and testing stages**

| Classification Technique | Best input features | Training     |                   | Testing      |                      |
|--------------------------|---------------------|--------------|-------------------|--------------|----------------------|
|                          |                     | Accuracy (%) | Training time (s) | Accuracy (%) | Prediction time (ms) |
| NB                       | WS, SC, WT          | 87.31        | 0.033             | 87.42        | 1.006                |
| DT                       | WS, SC, WT          | 99.46        | 0.040             | 98.69        | 1.009                |
| MLP                      | WS, SC, WT          | 98.93        | 39.649            | 98.40        | 127.241              |
| SVM                      | WS, SC, WT          | 97.66        | 0.315             | 96.67        | 155.508              |
| KNN                      | WS, SC, WT          | 98.79        | 0.045             | 96.95        | 80.848               |
| LR                       | WS, SC              | 91.78        | 0.115             | 90.14        | 1.001                |
| RF                       | WS, SC, WT          | 99.21        | 0.435             | 98.78        | 23.673               |
| XGB                      | WS, SC, WT          | 99.95        | 0.524             | 98.73        | 4.495                |
| AdaBoost                 | WS, SC              | 79.63        | 0.399             | 78.26        | 26.46                |

# Training performance of the implemented MLP algorithm

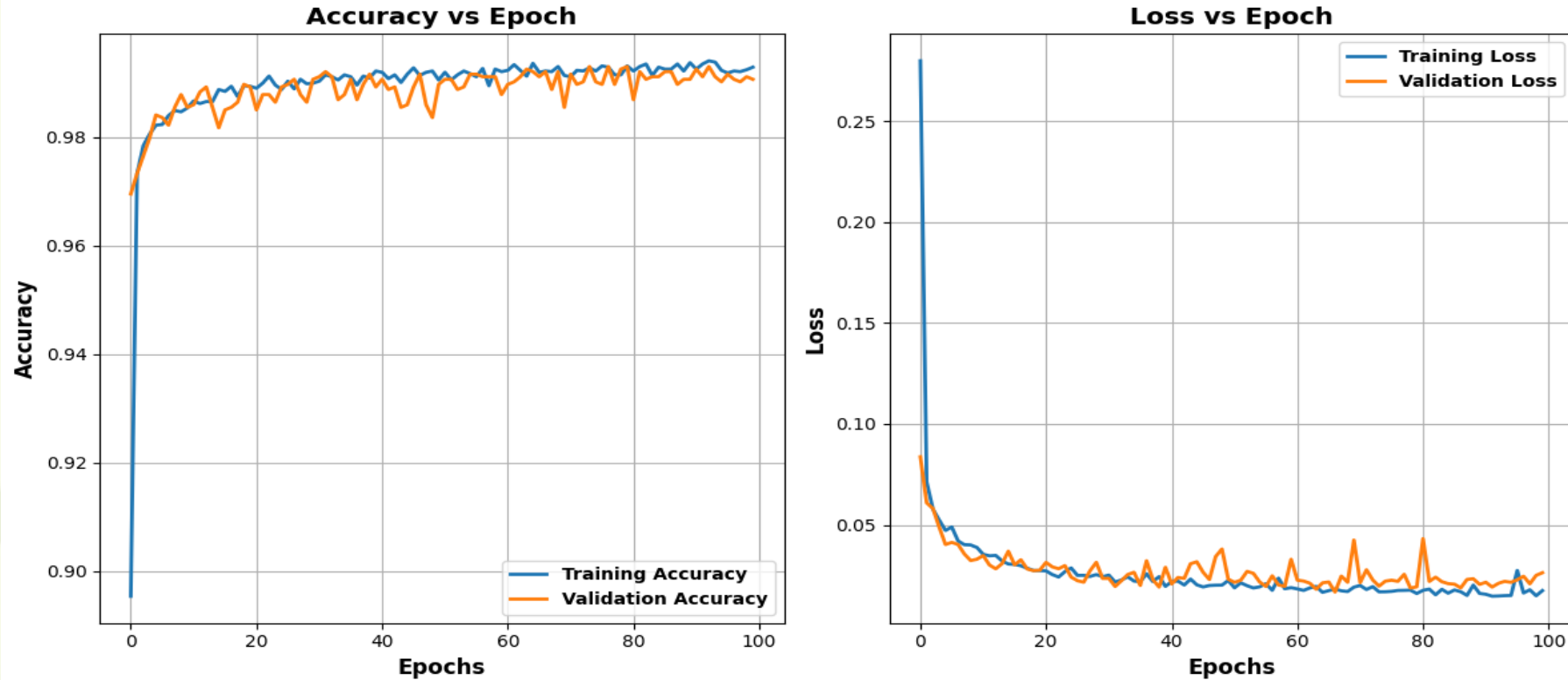


Fig. 13. Training performance of the implemented MLP algorithm (a) accuracy vs epoch (b) loss vs epoch

Table 8- MLP model performance evaluation

| Measurement Parameters | Non SSO | Non-Severe SSO | Severe SSO |
|------------------------|---------|----------------|------------|
| Precision              | 0.99    | 0.98           | 1          |
| Recall                 | 1       | 0.99           | 0.96       |
| F1-score               | 1       | 0.98           | 0.98       |

- The proposed SSO prediction model can predict the occurrence of SSO and send alarm signals within 314.842 ms. the prediction time of the MLP model itself is only 0.06 sec. However, the total time taken for prediction of a non-SSO, non-severe SSO, or severe SSO increases up to 157.142 ms. Existing ML-based research works required the analysis of data windows of size 1 section on top of their model prediction times for the prediction of SSO. Hence, the proposed methods can be suggested as a much faster approach for the prediction and alarming of SSO conditions. To understand the proposed SSO prediction algorithm, a further summary of the predicted class and the proposed actions is given in Table 9.

**Table 9- SSO model prediction and proposed control action taken**

| Prediction                | Proposed Action                       |
|---------------------------|---------------------------------------|
| Non SSO                   | No action needed                      |
| Non-Severe SSO            | Issue a warning signal                |
| Severe SSO                | Activation of SSO mitigation measures |
| Low confidence prediction | Alarms PSO for further investigations |

# Conclusion



- This study proposes an MLP algorithm to predict severe SSO conditions in DFIG-based WF connected to compensated lines that could trigger the wind generator protection devices.
- The IEEE's first benchmark model is used for dataset development.
- Three key factors affecting the severity of SSO: WS, SC, and WT are used as the attributes to train the proposed MLP classifier.
- The TA is used to determine the severity of the SSO.
- The MLP algorithm shows superior performance when compared with other ML algorithms. It has an accuracy of 99.01% and training and prediction times of 36.44 seconds and 0.06 sec.
- The total time taken for the proposed SSO prediction algorithm to predict the occurrence of severe SSO, non-severe SSO, and non-SSO is 157.142 ms.
- One of the key advantages of the proposed method over existing research is that this method uses an ML algorithm, which gives an idea regarding the confidence level of the prediction. This is crucial for modern power grid applications due to the dynamic nature of the power grid.

Integrated wind farm when compared to conventional synchronous generator under transient behavior



synchronous generators accelerate or decelerate



generator rotor speed and grid frequency changes



Inertia plays a role in suppressing the rapid changes in rotor speed and electrical power output



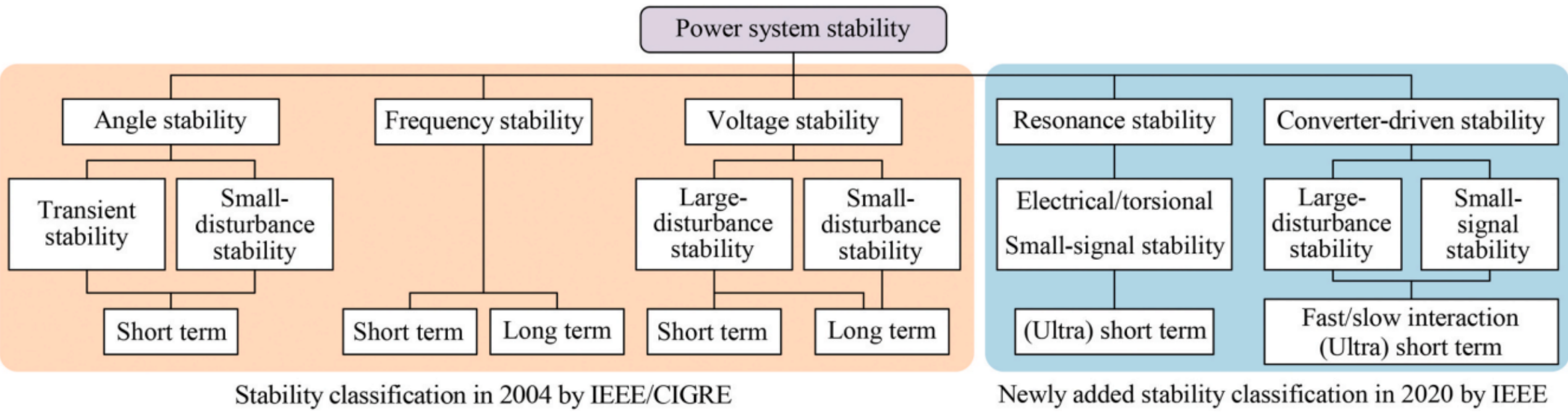
Wind farm are generally, back-to-back converters provide a fast and accurate variable-frequency excitation ( $\pm 30\%$ ) or fully rated converter



No concepts of “relative swing”, “rotor angle deviation”, “power angle characteristic”, and no inertia concept

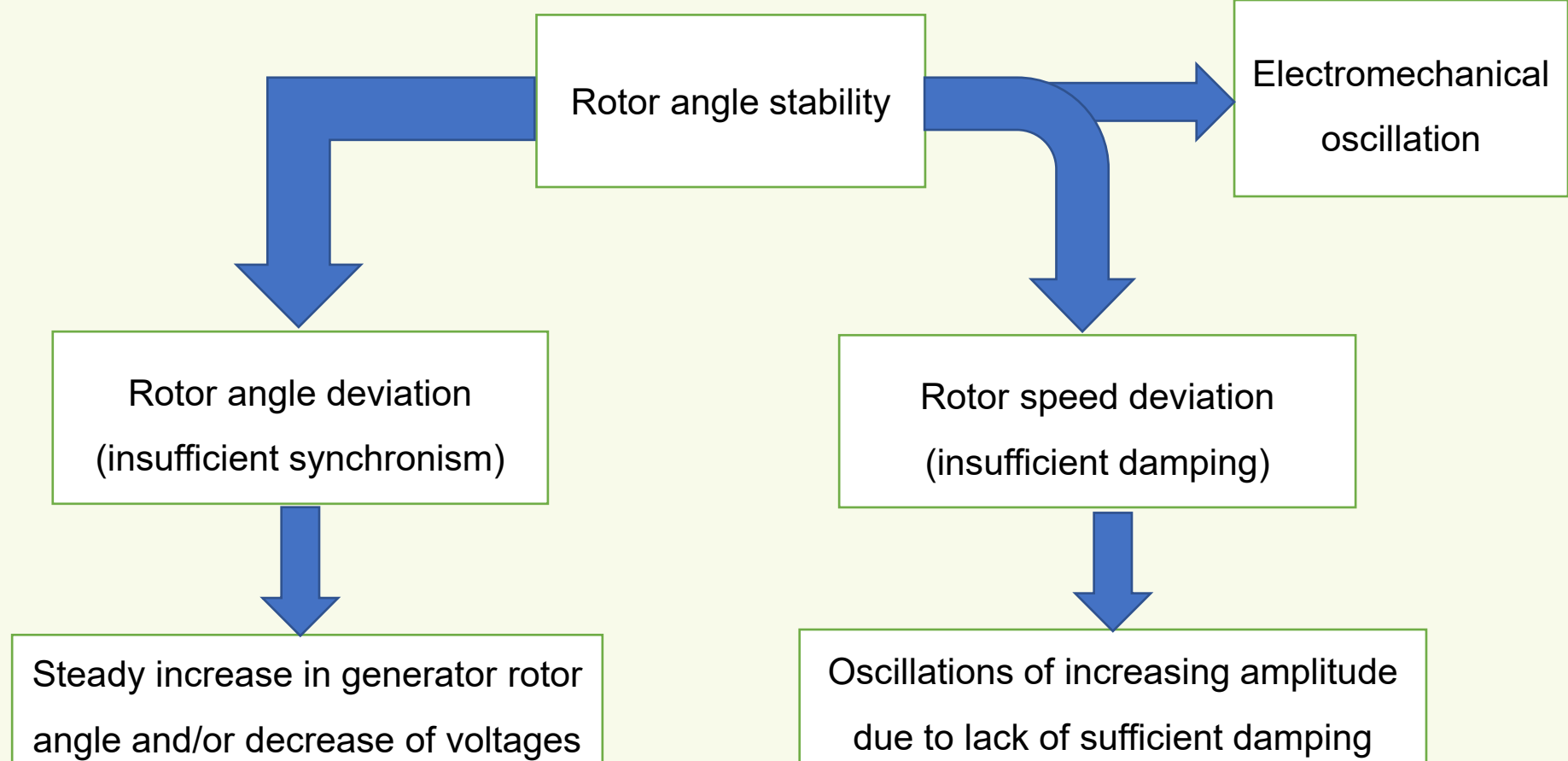


It suffers from Low Voltage Ride through (LVRT)



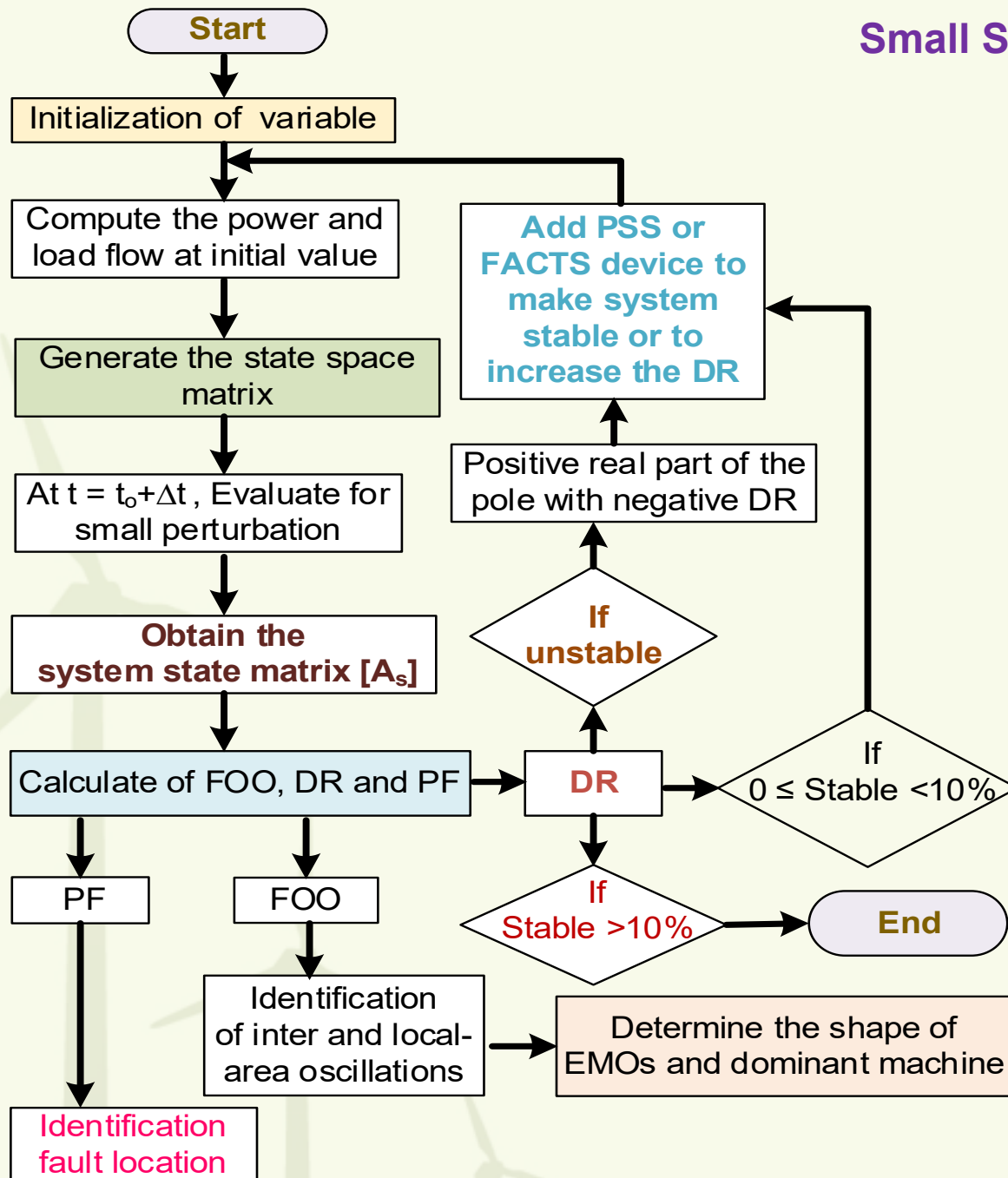
A reorganized version of the power system stability classification.

| Disturbance size  | Time scale                                  | Type of stability                                                                 |
|-------------------|---------------------------------------------|-----------------------------------------------------------------------------------|
| Large disturbance | Long term: tens of seconds to a few minutes | Long-term frequency stability, Large-disturbance long-term voltage stability      |
|                   | Short term: 0.1–2 Hz to 0.1–10 Hz           | Short-term frequency stability, Transient stability, Short-term voltage stability |
|                   | Short to ultra-short term: 1~kHz            | Large-disturbance converter driven stability                                      |
| Small disturbance | Long term: tens of seconds to a few minutes | Small-disturbance voltage stability                                               |
|                   | Short term: 0.1–2 Hz to 0.1–10 Hz           | Small-disturbance angle stability                                                 |
|                   | Short to ultra-short term: 1~kHz            | Resonance stability, Small-disturbance converter driven stability                 |



- Power system are related to the damping of the **electromechanical modes of oscillation**.
- **Local area mode of oscillation** means one generator swings against the rest of the generator at a frequency of 1.0 to 2.0 Hz.
- **Inter area mode of oscillation** means it involves two coherent group or groups of generators swinging against each other at a frequency of 1 Hz or less.

# Small Signal Stability Analysis



Flow chart of Small Signal stability Analysis

## Eigenvalues computation

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{f}(\mathbf{x}, \mathbf{y}, \mathbf{u}) \\ 0 &= \mathbf{g}(\mathbf{x}, \mathbf{y}, \mathbf{u}) \end{aligned}$$

## At the equilibrium point

$$\begin{aligned} 0 &= \mathbf{f}(\mathbf{x}_0, \mathbf{y}_0, \mathbf{u}_0) \\ 0 &= \mathbf{g}(\mathbf{x}_0, \mathbf{y}_0, \mathbf{u}_0) \\ \dot{\Delta \mathbf{x}} &= \mathbf{A} \Delta \mathbf{x} + \mathbf{B} \Delta \mathbf{u} \\ \Delta \mathbf{y} &= \mathbf{C} \Delta \mathbf{x} + \mathbf{D} \Delta \mathbf{u} \end{aligned}$$

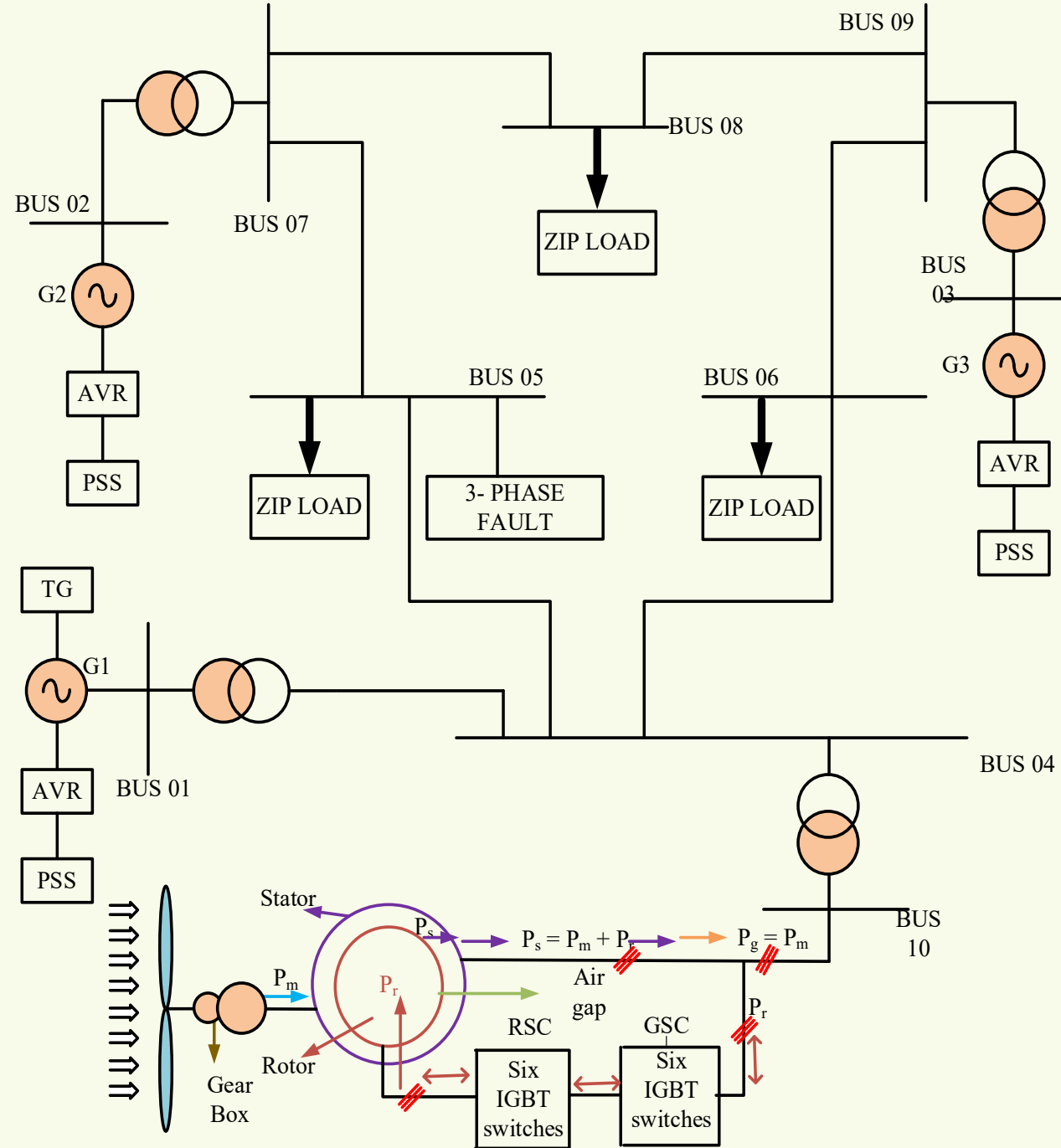
## Complex conjugate pairs

$$\lambda_i = \alpha_i \pm j\beta_i$$

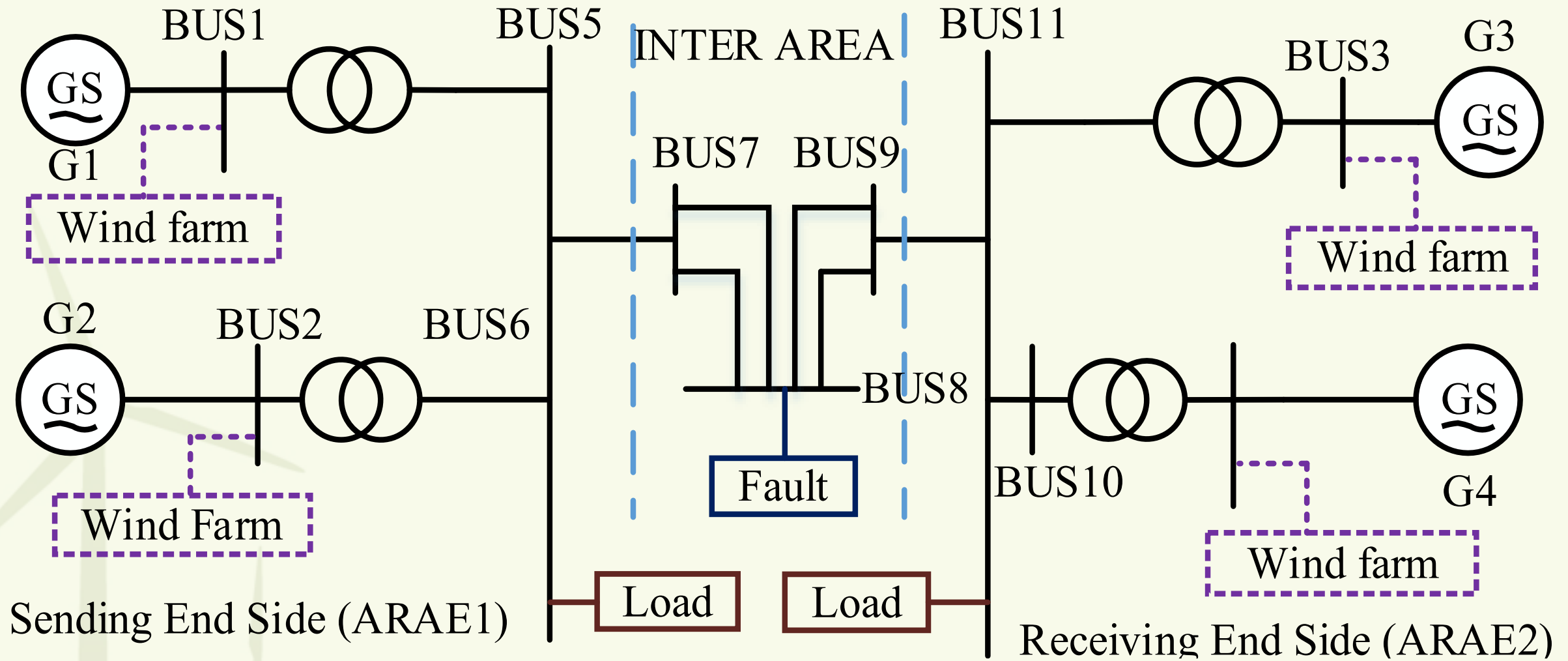
## DR, FOO and PF

$$\xi = \frac{-\alpha}{\sqrt{\alpha^2 + \beta^2}}, f = \frac{\beta}{2\pi}$$

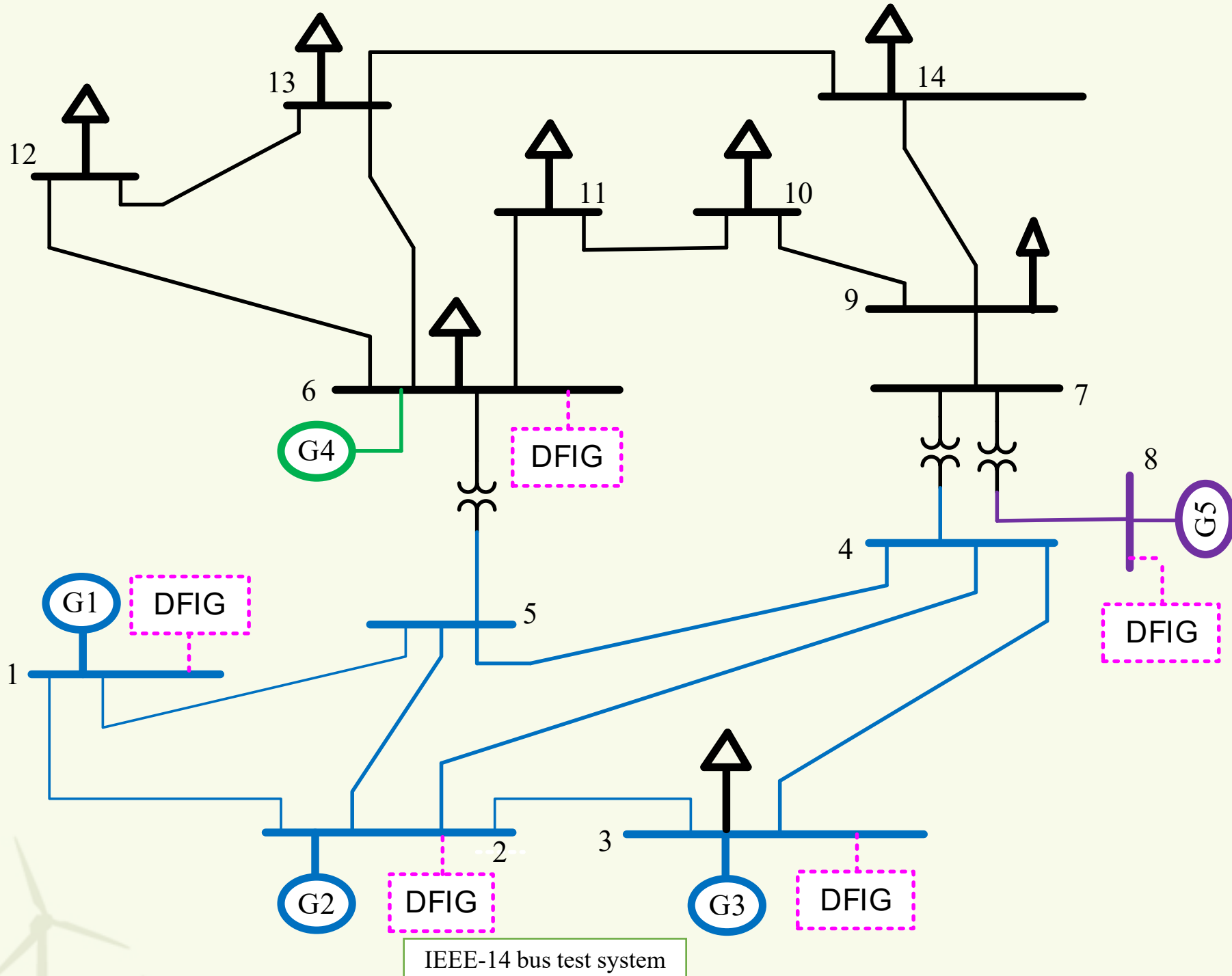
$$PF = \frac{|\phi_{mn}| |\psi_{nm}|}{\sum_{n=1}^k |\phi_{mn}| |\psi_{nm}|}$$



**Modified single line diagram of the IEEE-9 bus test system**



Two area test system (IEEE-11 Bus)



## Conclusions

- ❑ Brief review and comparison of the different WECS, various topologies are described, the machines involved in WECS ,weaknesses and strengths of different wind turbine generators are discussed.
- ❑ The topic research of power system stability with wind power penetration focuses on the modal analysis and time-domain simulations, through which the impact of wind power on power system oscillation modes and damping performances are discussed.



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